Restriction or misspecification? Dissecting the errors of classical and neural data assimilation

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First draft (structure v4) — October 15, 2026

Abstract

Model-based data assimilation (DA) is often outperformed by estimators that use less physics: observation-only products can beat ocean reanalyses, learned priors can beat the true dynamical model, and neural schemes are increasingly reported to beat classical ones. The usual explanations (a wrong model, a lossy algorithm, a poor tuning) are seldom separated. We cast every DA scheme, classical or learned, as an estimator of the conditional expectation of the state given its context, and represent posteriors with flow matching. The estimation error then decomposes along a chain of idealisations. The *irreducible* term is what no scheme can beat. A *restriction* is the cost of using less information than is available. A *misspecification* is the cost of believing a wrong model. An *approximation* is the cost of computing one's own target inexactly. Extending the chain to the full posterior yields a proper scoring rule that evaluates both the mean estimate and the uncertainty a scheme reports. On three case studies (a Lorenz-63 system, a two-scale Lorenz-96 system and a two-layer quasi-geostrophic ocean), we compare DA estimation errors and uncertainties along three axes: model-based versus neural DA schemes, perfect-model versus model-error settings, and model-based versus neural priors of the dynamics. Within a perfect-model setting, the apparent edge of neural schemes (30–45 % lower RMSE) is largely a restriction and approximation deficit of classical DA, and it shrinks with smoothing, larger ensembles and more optimisation iterations. Under model error, misspecification dominates: it appears in the low-noise score before it appears in the RMSE, it degrades calibration, and it makes the best perfect-model scheme (strong-constraint 4D-Var) the worst. Finally, a learned prior competes with the true-model hard constraint in perfect-model setting, but results in a much greater robustness to parameter tuning and forcing errors. For sparsely observed systems with significant model error, such as ocean reanalyses, our findings suggest that neural DA schemes can better reveal the information actually gained from a given observation configuration, and that progress lies less in more observations or a more faithful hard-constrained model than in representing model error explicitly and in assessing reanalyses with scores that resolve their reported uncertainty, not RMSE alone.

1 Introduction

Data assimilation (DA) combines a dynamical model with observations to estimate the state of a system [13]. Its premise is that the model carries information the observations lack, so that a model-based estimate should beat any estimate that ignores the model. The relevance of state-of-the-art data assimilation systems is widely acknowledged to support the monitoring and forecasting of earth system dynamics [4, 13, 29, 37].

Despite these successes, practice keeps contradicting the above premise, especially when dealing with observation-sparse context. As an example, gridded sea level anomaly products from observation-only optimal and neural interpolation resolves finer scales than an eddy-permitting ocean reanalysis [3, 25]. Over the recent years, numerous learned and hybrid schemes have been reported to outperform classical DA for both toy dynamical systems and real-wrold ones

[6, 10, 43]. Three explanations are usually offered: the dynamical model is wrong; the DA algorithm loses information (filtering instead of smoothing, Gaussian updates, finite ensembles, local optimisation); or the scheme does not use all the information available to it. They are rarely separated, because each scheme mixes all three, and a comparison through classic estimation evaluation metrics (e.g., MSE, CRPS,...) cannot tell them apart.

This study aims to disentangle these different contributions in the diagnosis of the overall error budget as well as to understand to which extent this diagnosis is method-specific or tied to the intrinsic properties of the considered dynamics and observing system. Formally, we treat every DA scheme, classical or learned, as an estimator of the conditional distribution $p(\mathbf{x} \mid C)$ of the state $\mathbf{x}$ given a context C, and of its mean $\mathbb{E}[\mathbf{x} \mid C]$. The error of a scheme classically splits along a chain of idealisations: an irreducible term; a *restriction*, because the scheme uses less information than C; a *misspecification*, because it believes a model other than the true one; and an *approximation*, because it computes its own target only approximately. Two schemes with the same RMSE can have very different budgets, and only the budget says what to fix. Through a flow matching representation of posteriors, we show the same chain applies to the uncertainty a scheme reports. Overall, the proposed approach allows us to assess the extent to which the resulting scores inform the true underlying posteriors $p(\mathbf{x} \mid C)$ or reflect the impact of DA models' and algorithms' errors and assumptions. Through experiments on three case studies (a Lorenz-63 system, a two-scale Lorenz-96 system and a two-layer quasi-geostrophic ocean), we diagnose the error budget of model-based and neural DA schemes under perfect-model and model-error settings and investigate whether the form of the prior, derived from the model equations or learned from data, drives the performance.

This paper is organized as follows. Section 2 introduces some background on Bayesian data assimilation. Section 3 presents the proposed flow matching framework to diagnose the estimation performance of data assimilation schemes. Section 4 details the considered experimental design and we present our results in Section 5. We discuss further our findings in Section 6

2 Background

2.1 Bayesian data assimilation

Let $\mathbf{x} = (\mathbf{x}_0, \dots, \mathbf{x}_K)$ be the state trajectory over an assimilation window, $\mathbf{z}$ an external forcing and $\boldsymbol{\theta}$ the physical parameters. The discrete dynamics are given by

$$\mathbf{x}_{k+1} = \mathcal{M}_\Delta(\mathbf{x}_k, \mathbf{z}_k; \boldsymbol{\theta}) + \xi_k, \tag{1}$$

with $\mathcal{M}_\Delta$ the time-stepping operator over a physical step Δ and ξ_k a model-error process; the observations are

$$\mathbf{y}_k = \mathcal{H}(\mathbf{x}_k) + \boldsymbol{\epsilon}_k, \qquad \boldsymbol{\epsilon}_k \sim \mathcal{N}(0, \mathbf{R}), \tag{2}$$

at the observation times of the window. Everything an estimator may condition on is collected in the *context* $C = (\mathbf{y}, \boldsymbol{\theta}, \mathbf{z})$. The posterior

$$p(\mathbf{x} \mid C) \;\propto\; p(\mathbf{y} \mid \mathbf{x})\, p(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{z}) \tag{3}$$

combines the observation likelihood (through $\mathcal{H}$ and $\mathbf{R}$) with a prior that, in model-based DA, is generated by the dynamics (1). Learned schemes often replace or bypass this prior. Here, we focus on state estimation problems, meaning that $\boldsymbol{\theta}$ and $\mathbf{z}$ are inputs that a scheme may use. We discuss extending this approach to parameter estimation in Section 6. The conditional expectation $m = \mathbb{E}[\mathbf{x} \mid C]$ is the associated MSE-optimal estimator, meaning that $\mathbb{E}[\|\mathbf{x} - \mathbb{E}[\mathbf{x} \mid C]\|^2]$ is the smallest mean-squared error any scheme can reach.

We can distinguish two main categories of model-based DA schemes to address the estimation of $\mathbb{E}[\mathbf{x} \mid C]$ or of the full posterior $p(\mathbf{x} \mid C)$, namely variational (4D-Var) DA schemes and ensemble Kalman ones [13, 15]:

- **4D-Var schemes:** they compute the maximum a posteriori (MAP) trajectory of (3) under a Gaussian prior $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{x}^b, \mathbf{B})$ on the initial state and the Gaussian observation model (2). Minimising $-\log p(\mathbf{x} \mid C)$ amounts to minimising

$$J(\mathbf{x}) = \tfrac{1}{2}\|\mathbf{x}_0 - \mathbf{x}^b\|^2_{\mathbf{B}^{-1}} + \tfrac{1}{2}\sum_k \|\mathbf{y}_k - \mathcal{H}(\mathbf{x}_k)\|^2_{\mathbf{R}^{-1}} + \tfrac{1}{2}\sum_{k=0}^{K-1} \|\mathbf{x}_{k+1} - \mathcal{M}_\Delta(\mathbf{x}_k, \mathbf{z}_k; \boldsymbol{\theta})\|^2_{\mathbf{Q}^{-1}}, \quad (4)$$

  where $\|\mathbf{u}\|^2_{\mathbf{A}} = \mathbf{u}^\top \mathbf{A}\, \mathbf{u}$. In the *strong-constraint* form, the model is assumed perfect ($\mathbf{Q} \to 0$): the trajectory is a deterministic function of $\mathbf{x}_0$, $\mathbf{x}_k = \mathcal{M}^k_\Delta(\mathbf{x}_0)$, and the minimisation is performed over $\mathbf{x}_0$ only. In the *weak-constraint* form, the model-error process $\xi_k \sim \mathcal{N}(0, \mathbf{Q})$ of (1) is retained and the whole trajectory is optimised. The gradient of J is obtained with the adjoint of $\mathcal{M}_\Delta$ and J is minimised iteratively, typically with a quasi-Newton method such as L-BFGS. Long windows are usually split into sub-windows that are cycled, the analysis of one providing the background of the next. 4D-Var returns the mode of the posterior, which coincides with the conditional mean m only when the posterior is symeetric, that is for instance the case linear-Gaussian dynamics and observation operators. Hereafter, we will consider both strong and weak 4DVar schemes, respectively referred to as S-4DVar and W-4DVar.

- **Kalman schemes:** they process observations sequentially. Assuming Gaussian errors, the filter propagates the mean and covariance of $p(\mathbf{x}_k \mid \mathbf{y}_{\le k})$. The forecast step $\mathbf{x}^f_k = \mathcal{M}_\Delta(\mathbf{x}^a_{k-1}, \mathbf{z}_{k-1}; \boldsymbol{\theta})$, $\mathbf{P}^f_k = \mathbf{M}_k \mathbf{P}^a_{k-1} \mathbf{M}_k^\top + \mathbf{Q}$ is followed by the analysis step

$$\mathbf{x}^a_k = \mathbf{x}^f_k + \mathbf{K}_k\big(\mathbf{y}_k - \mathcal{H}(\mathbf{x}^f_k)\big), \quad \mathbf{K}_k = \mathbf{P}^f_k \mathbf{H}_k^\top \big(\mathbf{H}_k \mathbf{P}^f_k \mathbf{H}_k^\top + \mathbf{R}\big)^{-1}, \quad \mathbf{P}^a_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)\, \mathbf{P}^f_k, \quad (5)$$

  where $\mathbf{M}_k$ and $\mathbf{H}_k$ are the linearisations of $\mathcal{M}_\Delta$ and $\mathcal{H}$. This is the exact posterior for linear Gaussian systems. The *ensemble Kalman filter* (EnKF) [13, 21] replaces $(\mathbf{x}^f_k, \mathbf{P}^f_k)$ by the empirical mean and covariance of an ensemble of N members propagated with the nonlinear model, which avoids both the linearisation and the storage of covariance matrices. The analysis update is applied to each member, either with perturbed observations (stochastic EnKF) or through a deterministic square-root transform of the ensemble anomalies (ETKF). It is usually complemented by covariance inflation, to compensate for sampling errors and model error, and by localisation. Ensemble smoothers (EnKS, ETKS) apply the same update to a whole window, so that the state at each time is conditioned on all observations of the window and not only on the past ones. The ensemble provides samples that are used as an approximation of the posterior, with the Gaussian assumption in the analysis step. Hereafter, we will focus on ETKF, EnKF and ETKS schemes as representative of the Kalman DA category.

Regarding neural DA schemes, we can distinguish deterministic end-to-end schemes aiming at approximating $\mathbb{E}[\mathbf{x} \mid C]$ and probabilistic approaches aiming at approximating or sampling the full posterior $p(\mathbf{x} \mid C)$:

- **Deterministic neural schemes** train a neural operator Ψ_Θ that maps the context C to a state estimate, such that

$$\forall C, \quad \Psi_{\Theta^*}(C) \approx \mathbb{E}[\mathbf{x} \mid C]. \quad (6)$$

  The conditional mean being the MSE-optimal estimator, the associated training loss is the mean squared error over training pairs $(\mathbf{x}, C)$,

$$\Theta^* = \arg\min_\Theta \; \mathbb{E}\big[\|\Psi_\Theta(C) - \mathbf{x}\|^2\big], \quad (7)$$

  where the expectation is taken over the joint distribution of states and contexts, hence over the observing systems, parameters and forcings seen at training. Once trained, a single

forward pass replaces the iterative minimisation of 4D-Var or the ensemble propagation of Kalman schemes. U-Net and transformer architectures are state-of-the-art parametric families classically used for reconstruction problems on n-dimensional tensors [41]. The operator only sees what it was trained on: if some components of C (for example $\boldsymbol{\theta}$ or $\mathbf{z}$) are not provided as inputs, the network learns the conditional mean marginalised over them, and the estimate degrades when the observing system leaves the training distribution. Hereafter, we will focus on UNet architctures and refer to this subcategory of neural schemes as M-Unet.

- **Probabilistic neural schemes.** use generative models, such as diffusion models [44] and flow matching [35], to sample the posterior rather than to estimate its mean. Using the linear path (11) between a noise sample $\mathbf{x}_0 \sim \mathcal{N}(0, I)$ and a target state $\mathbf{x}_1$, conditional flow matching learns the velocity field $v_\Theta(\mathbf{x}_\tau, \tau, C)$ by regression,

$$\Theta^* = \arg\min_\Theta \mathbb{E}\Big[\big\|v_\Theta(\mathbf{x}_\tau, \tau, C) - (\mathbf{x}_1 - \mathbf{x}_0)\big\|^2\Big], \tag{8}$$

whose minimiser is $v(\mathbf{x}_\tau, \tau, C) = \mathbb{E}[\mathbf{x}_1 - \mathbf{x}_0 \mid \mathbf{x}_\tau, C] = \big(\Psi(\mathbf{x}_\tau, \tau, C) - \mathbf{x}_\tau\big)/(1-\tau)$, with $\Psi = \mathbb{E}[\mathbf{x}_1 \mid \mathbf{x}_\tau, C]$ the denoiser of the posterior at noise level τ. Samples of $p(\mathbf{x} \mid C)$ are then generated by integrating the ODE $\dot{\mathbf{x}}_\tau = v_\Theta(\mathbf{x}_\tau, \tau, C)$ from $\tau = 0$ to $\tau = 1$, for instance with a few Euler steps, and an ensemble is obtained by repeating the draw from different noise samples. At $\tau = 0$, Ψ reduces to the conditional mean (6), so that the deterministic schemes are the mean part of this representation and the flow adds the uncertainty on top of it. This conditional setting, in which the full context C is an input of the network, is the direct counterpart of the deterministic scheme above. Hereafter, we will focus on this conditional flow matching referred to as CFM. A second route does not condition on the observations at training time. The generative model is trained to represent the dynamical prior $p(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{z})$ only, and the posterior (3) is obtained at inference by *observation-guided sampling* [16, 43, 19]: the likelihood term is injected into the sampler through the known observation model (2) similarly to model-based DA schemes. Because the guided likelihood is only approximate (Gaussian approximation, tempering), the resulting samples approximate the true posterior only up to a structural error. Following [43], we will refer to this class of neural approaches as score-based data assimilation schemes (SDA).

2.2 The estimation error chain for $\mathbb{E}[\mathbf{x} \mid C]$

Let us focus here on the skills of a DA scheme in terms of estimation of the conditional mean $m_C = \mathbb{E}[\mathbf{x} \mid C]$. The DA scheme departs from this MSE-optimal estimation in three ways:

- the DA scheme may use some information $S \subseteq C$ rather than the full context C: for instance, contrary to a smoother, a DA filter uses the observations up to the current time; a learned DA scheme may ignore $\boldsymbol{\theta}$ and $\mathbf{z}$;
- It believes a model $\tilde{p}(\cdot|C)$ that may differ from the true posterior $p(\cdot|C)$: for instance, 4DVar schemes relate to considering the MAP estimate and Ensemble Kalman methods exploit Gaussian approximation, which can be interpreted as not considering the true posterior p;
- It computes its own target only approximately: this accounts among other for the number of optimization steps, finite-ensemble approximations, training dataset size...

This leads to the corresponding chain of estimators as

$$m = \mathbb{E}[\mathbf{x} \mid C] \;\longrightarrow\; m_S = \mathbb{E}[\mathbf{x} \mid S] \;\longrightarrow\; \tilde{m} = \mathbb{E}_{\tilde{p}}[\mathbf{x} \mid S] \;\longrightarrow\; \hat{m}, \tag{9}$$

Table 1: **Mapping the design choices of DA schemes onto the error chain (9).**

(a) Model-based DA.

Design choice	error chain component	schemes concerned
filtering ($S = \mathbf{y}_{t' \leq t}$)	restriction	EnKF, ETKF
cycled sub-windows	restriction	S-4D-Var, W-4DVar
model error in $\mathcal{M}$, $\boldsymbol{\theta}$, $\mathbf{z}$	misspecification	all
Gaussian update	misspecification	EnKF, ETKF, ETKS
MAP instead of conditional mean	misspecification	S-4D-Var, W-4DVar
finite ensemble	approximation	EnKF, ETKF, ETKS
non-convex/local optimization	approximation	S-4D-Var, W-4DVar
finite optimization step number	approximation	S-4D-Var, W-4DVar

(b) Neural DA.

design choice	link	schemes concerned
ignoring $\boldsymbol{\theta}$, $\mathbf{z}$ as inputs	restriction	all
out-of-distribution inference	misspecification	all
observation-guided sampling	misspecification	SDA
finite network	approximation	all
finite training dataset	approximation	all

where $\hat{m}$ is what the DA scheme outputs. Assuming that estimate $\hat{m}$ depends only on subcontext S and on randomness independent of the truth, we can derive the following decomposition of the associated mean square estimation error $\mathbb{E}\|\hat{m} - \mathbf{x}\|^2$

$$\mathbb{E}\|\hat{m} - \mathbf{x}\|^2 = \underbrace{\mathbb{E}\,\mathrm{tr}\,\mathrm{Cov}(\mathbf{x} \mid C)}_{\text{irreducible}} + \underbrace{\mathbb{E}\|m - m_S\|^2}_{\text{restriction}} + \underbrace{\mathbb{E}\|m_S - \hat{m}\|^2}_{\text{misspecification+approximation}} \tag{10}$$

The first holds because m is the conditional expectation given C. The second holds because $m_S = \mathbb{E}[m \mid S]$ (tower property). The third holds because $m_S - \tilde{m}$ is a function of S and of independent randomness only, so the cross terms vanish. The latter also applies to the fourth term. One cannot in general split the third term into distinct mispecification and approximation terms, as their conditional independence w.r.t. S is not always verified.

We can analyse the design choices in the above-mentioned model-based and neural schemes according to the associated error term as summarized in Table 1. These features are choices, not errors: each buys something — tractability, a sequential implementation, robustness to the observing system — and the question is what it costs, and when the cost changes nature. This synthesis emphasizes that all model-based DA schemes involve a non-zero mispecification error. By contrast, end-to-end neural schemes (M-Unet, CFM) may just involve approximation error terms given their versatility to embed any type of conditioning and universal approximation theorem for state-of-the-art neural network families [30, 18]. This likely provides first insights to support why neural schemes can outperform model-based DA schemes when dealing with in-distribution generalization [22, 9], *i.e.* when the test data fall within the same distribution as the training data, which is usually the case in idealized evaluation benchmark [10, 22, 43]. Beyond this idealized configuration, one can obviously question how the mispecification term grows for neural schemes under out-of-distribution inference compared to that of model-based DA ones.

3 Scoring the uncertainty using flow matching

We extend the chain (10) scores for the mean or deterministic state estimate of a scheme to the uncertainty component – an ensemble spread, a Gaussian covariance, a set of samples – report by probabislitic DA schemes. We exploit a flow-matching representation of any posterior and introduce a family of strictly proper scores, indexed by $\tau \in (0,1)$, each of which characterizes the posterior uniquely and splits along the restriction–misspecification–approximation chain; at $\tau = 0$ it reduces to the mean-squared error of the mean.

3.1 Flow matching representation of a posterior

Let us consider the classic notations used in the flow matching literature [35, 2] where $\mathbf{x}_1$ denotes a centered and standardized target state (a draw from the scheme's output, or the truth $\mathbf{x}_1^\star$). Flow matching methods consider a linear path between the target space and a Gaussian space:

$$\mathbf{x}_\tau = \tau\,\mathbf{x}_1 + (1-\tau)\,\mathbf{x}_0, \qquad \mathbf{x}_0 \sim \mathcal{N}(0, I), \quad \tau \in [0,1), \tag{11}$$

Any distribution $q(\mathbf{x}_1 \mid C)$ of a scheme uniquely defines the operator

$$\Psi_q(\mathbf{x}_\tau, C, \tau) \;=\; \mathbb{E}_q[\mathbf{x}_1 \mid \mathbf{x}_\tau, C] \tag{12}$$

such that integrating the ODE $\dot{\mathbf{x}}_\tau = (\Psi(\mathbf{x}_\tau, \tau, C) - \mathbf{x}_\tau)/(1-\tau)$ between $\tau = 0$ to $\tau = 1$ from Gaussian states $\mathbf{x}_0$ generates samples according to posterior $q(\mathbf{x}_1 \mid C)$.

3.2 The flow-matching score and the chain at every τ

We define the flow-matching score of a scheme, characterized by its flow Ψ_q as the expectation of the quadratic error of its operator against the reference distribution p for any $\tau \in (0,1)$

$$FMS_\tau(p \,\|\, q) \;=\; \mathbb{E}\big\|\Psi_q(\mathbf{x}_\tau, \tau) - \mathbf{x}_1)\big\|^2, \qquad \mathbf{x}_\tau = \tau\,\mathbf{x}_1 + (1-\tau)\,\mathbf{x}_0.\ \mathbf{x}_1 \sim p,\ \mathbf{x}_0 \sim \mathcal{N}(0, I) \tag{13}$$

The score is related to the divergence between the operators of q and p,

$$D_\tau(p \,\|\, q) = \mathbb{E}\big\|\Psi_q(\mathbf{x}_\tau, C, \tau) - \Psi_p(\mathbf{x}_\tau, C, \tau)\big\|^2, \tag{14}$$

through the exact decomposition

$$\mathrm{FMS}_\tau(q) = D_\tau(p \,\|\, q) + \mathrm{mmse}_p(\tau), \qquad \mathrm{mmse}_p(\tau) = \mathbb{E}\big\|\Psi_p(\mathbf{x}_\tau, C, \tau) - \mathbf{x}_1^\star\big\|^2, \tag{15}$$

which holds because $\Psi_p = \mathbb{E}[\mathbf{x}_1^\star \mid \mathbf{x}_\tau, C]$, so that $\mathbf{x}_1^\star - \Psi_p$ is orthogonal to any function of $(\mathbf{x}_\tau, C)$. The term $\mathrm{mmse}_p(\tau)$ is the irreducible error on $\mathbf{x}_1$ given $\mathbf{x}_\tau$ and C. It does not depend on q. By Tweedie's formula [20, 40], $\tau\,\Psi_\rho(\mathbf{x}_\tau, C, \tau) = \mathbf{x}_\tau + (1-\tau)^2\,\nabla \log \rho_\tau(\mathbf{x}_\tau \mid C)$ for any distribution ρ, where ρ_τ is the law of $\mathbf{x}_\tau$ when $\mathbf{x}_1 \sim \rho(\cdot \mid C)$. Hence, the considered divergence is a scaled Fisher divergence between the noised posteriors p_τ and q_τ, the denoising form of score matching [32, 48].

$$D_\tau(p \,\|\, q) = \frac{(1-\tau)^4}{\tau^2}\,\mathbb{E}_C\,\mathbb{E}_{\mathbf{x}_\tau \sim p_\tau(\cdot|C)}\big\|\nabla \log p_\tau(\mathbf{x}_\tau \mid C) - \nabla \log q_\tau(\mathbf{x}_\tau \mid C)\big\|^2, \tag{16}$$

The considered score has the following properties:

(i) At $\tau = 0$ it is the mean-squared error of the scheme's mean.

(ii) For $\tau \in (0,1)$ it is strictly proper [28]: its expectation is minimised only by the true conditional distribution, $q(\cdot \mid C) = p(\cdot \mid C)$ for almost every context. In practice, as detailed for the ensemble form of the score (see below), the score is computed on one-dimensional slices $\mathbf{u}^\top \mathbf{x}_1$ of the state — the coordinate axes for the per-element evaluation, random directions for the sliced one. The computed score is then strictly proper for the distributions of the slices considered, i.e. for the marginal distributions in the per-element case; with directions drawn at random over a patch, its expectation over directions is strictly proper for the joint distribution of the patch (Cramér–Wold theorem).

(iii) For a Gaussian $\mathcal{N}(m, v)$ with error e, $\mathrm{FMS}_\tau = (e^2 + \alpha_\tau\, v^2)/(1 + \alpha_\tau\, v)^2$, minimised at $v = \mathbb{E}e^2$ for every τ: the spread–skill identity.

(iv) A purely-deterministic estimate leads to a constant score at every τ given by the mean-square error of the deterministic estimate.

Besides the decomposition between the mean at $\tau = 0$ and the distribution of the anomaly characterized by $\tau > 0$, the score at small τ characterize the space-time scale with the largest energy, in general large-scale patterns, and large τ reads the lower-energy components, in general fine-scale patterns [39, 33].

As the considered score relies on conditional expectations of the state, we can derive a decomposition similar to (10) for conditioning variables $(\mathbf{x}_\tau, C)$ instead of C only. Under the condition of (10), for every τ,

$$\mathrm{FMS}_\tau(q) \;=\; \underbrace{\mathrm{mmse}_p(\tau)}_{\text{irreducible}} \;+\; \underbrace{D_\tau(p \,\|\, p_S)}_{\text{restriction}} \;+\; \underbrace{D_\tau(p_S \,\|\, q)}_{\text{misspecification+approximation}}\;, \tag{17}$$

with equality if and only if $q = p$. At $\tau = 0$, (17) reduces to (10). Three consequences follow.

- **Differences are exact.** The irreductible component $\mathrm{mmse}_p(\tau)$ is unknown but common to all schemes, so score differences between schemes are exact differences of operator divergences. This also means that the lowest score among different schemes deliver an upper bound for the irreductible term;

- **A restriction identity along τ.** Consider two schemes using nested information $S' \subseteq S \subseteq C$, for instance a filter (S', observations up to the current time) and the smoother computed on the same ensemble (S, all observations of the window). If both were exact, $q' = p_{S'}$ and $q = p_S$, the chain (17) would give, at every τ,

$$\mathrm{FMS}_\tau(q') - \mathrm{FMS}_\tau(q) = D_\tau(p_S \,\|\, p_{S'}) = \mathbb{E}\big\|\Psi_q(\mathbf{x}_\tau, C, \tau) - \Psi_{q'}(\mathbf{x}_\tau, C, \tau)\big\|^2 \;\geq\; 0 : \tag{18}$$

  the score gap equals the distance between the two operators, which is computable without the truth. A gap that does not match this distance flags misspecification or approximation. At $\tau = 0$ the identity reads

$$\mathbb{E}\|\mathbf{m}_{q'} - \mathbf{x}_1^\star\|^2 - \mathbb{E}\|\mathbf{m}_q - \mathbf{x}_1^\star\|^2 = \mathbb{E}\,\mathrm{tr}\,\mathrm{Cov}_{q'}(\mathbf{x}_1 \mid S') - \mathbb{E}\,\mathrm{tr}\,\mathrm{Cov}_q(\mathbf{x}_1 \mid S) = \mathbb{E}\|\mathbf{m}_q - \mathbf{m}_{q'}\|^2, \tag{19}$$

  relating the MSE gap, the gap of reported variances and the mean shift between the two schemes, which we test in Section 5.1.

- **Restriction vs. misspecification and approximation.** For exact schemes with nested information, the score gap (18) is non-negative at every τ: a pure restriction raises the score curve uniformly in sign, and the less informed scheme stays calibrated for its own information. A gap that changes sign along τ therefore reveals misspecification or approximation, with a scale-dependent signature; for schemes with non-nested information, sign changes can also arise from restriction alone.

The chain decomposition is not unique to FMS: the expected log score splits the same way, through Kullback–Leibler divergences, and calibration–resolution decompositions exist for every proper score [11, 28]. We use FMS_τ for three reasons:

- it is computable for samples, Gaussians and point estimates alike, where the log score is undefined for ensembles and point masses;

- it is resolved over τ, which separates the mean from the uncertainty;

- it equals the MSE at $\tau = 0$, which keeps continuity with DA practice.

The CRPS remains a useful marginal check and is reported alongside.

3.3 Derived diagnostics

All diagnostics are computed per element (window, time step, observed channel) in standardised units and then averaged; m and v denote the mean and variance reported by the scheme, and $\bar{\cdot}$ the average over windows, times, channels and draws of $\mathbf{x}_0$.

Ensemble form of the score. When considering an ensemble $\{\mathbf{x}_1^{(i)}\}_{i=1}^N$ smoothed by a Gaussian kernel of variance h^2 (with $h^2 = 0$ for the empirical ensemble) to represent posterior $q(\mathbf{x}_1|C)$, the associated flow matching operator $\tilde{\Psi}_q$ is given by

$$\tilde{\Psi}_q(\mathbf{x}_\tau, C, \tau) = \sum_{i=1}^{N} w_i \Big[\mathbf{x}_1^{(i)} + \frac{\tau h^2}{c^2}\big(\mathbf{x}_\tau - \tau \mathbf{x}_1^{(i)}\big)\Big], \text{ with } w_i \propto \exp\Big(-\frac{\|\mathbf{x}_\tau - \tau \mathbf{x}_1^{(i)}\|^2}{2c^2}\Big), \tag{20}$$

with $c^2 = \tau^2 h^2 + (1-\tau)^2$. The ensemble form of the score, denoted $\tilde{\mathrm{FMS}}_\tau$, is the empirical average of $\|\tilde{\Psi}(\mathbf{x}_\tau, C, \tau) - \mathbf{x}_1\|^2$ over draws of $\mathbf{x}_1 \sim p$ and $\mathbf{x}_0 \sim \mathcal{N}(0, I)$. At $\tau = 0$, we retrieve the squared error of the ensemble mean. when $\tau \to 1$, the score converges to the squared error of the closest ensemble member for $h = 0$ and to 0 otherwise.

We exploit this ensemble form of the score as a diagnostic to intercompare different schemes and assess the underlying error budget decomposition. To retain the dependence between components without the degeneration of the joint ensemble operator, we evaluate the score on one-dimensional random projections $\mathbf{u}^\top \mathbf{x}_1$ of space–time patches, with $\mathbf{u}$ uniform on the unit sphere in standardised units, in the spirit of sliced comparisons of distributions [38, 8]. Since $\mathbf{u}^\top \mathbf{x}_0 \sim \mathcal{N}(0,1)$, each projection defines a valid one-dimensional score, and by the Cramér–Wold theorem [17] the expected score over directions is strictly proper for the joint distribution of the patch. Directions and noise draws are common random numbers: they are drawn per test item from a seed fixed by the item, and shared by all schemes, so that score differences between schemes are evaluated on identical draws.

Gaussian form of the score. Many DA schemes report, for each context C, a Gaussian approximation of the posterior,

$$q(\mathbf{x}_1 \mid C) = \mathcal{N}\big(\mathbf{m}_q(C),\, \mathbf{\Sigma}_q(C)\big), \tag{21}$$

where the mean $\mathbf{m}_q(C)$ and the covariance $\mathbf{\Sigma}_q(C)$ are context-dependent: they are computed from the information $S \subseteq C$ that the scheme uses (observations, and for model-based schemes parameters and forcing), for instance as the mean and covariance of an analysis ensemble. On the path (11), $(\mathbf{x}_1, \mathbf{x}_\tau)$ is jointly Gaussian given C, and the operator (12) is linear in $\mathbf{x}_\tau$,

$$\Psi_q(\mathbf{x}_\tau, C, \tau) = \mathbf{m}_q(C) + \mathbf{K}_\tau(C)\Big(\frac{\mathbf{x}_\tau}{\tau} - \mathbf{m}_q(C)\Big), \qquad \mathbf{K}_\tau(C) = \alpha_\tau\, \mathbf{\Sigma}_q(C)\big(\mathbf{I} + \alpha_\tau\, \mathbf{\Sigma}_q(C)\big)^{-1}, \tag{22}$$

with $\alpha_\tau = \tau^2/(1-\tau)^2$. Writing $\mathbf{e}(C) = \mathbf{m}_q(C) - \mathbf{x}_1^\star$ for the error w.r.t. the reported mean the resulting score in closed form is

$$\mathrm{FMS}_\tau^{\mathcal{G}}(\mathbf{m}_q, \mathbf{\Sigma}_q) = \mathbb{E}\Big[\mathbf{e}(C)^\top \big(\mathbf{I} + \alpha_\tau\, \mathbf{\Sigma}_q(C)\big)^{-2} \mathbf{e}(C) \;+\; \alpha_\tau\, \mathrm{tr}\Big(\mathbf{\Sigma}_q^2(C)\, \big(\mathbf{I} + \alpha_\tau\, \mathbf{\Sigma}_q(C)\big)^{-2}\Big)\Big], \tag{23}$$

where the expectation runs over contexts and truths. Considering the eigendecomposition of the covariance matrix, *i.e.* $\mathbf{\Sigma}_q(C) = \mathbf{U}(C)\, \mathbf{\Lambda}(C)\, \mathbf{U}(C)^\top$, with $\mathbf{\Lambda} = \mathrm{diag}(\{\lambda_1, \ldots, \lambda_n\})$ and $\tilde{\mathbf{e}}(C) = \mathbf{U}^\top \mathbf{e}(C)$, the score is a sum of univariate terms in the eigenbasis,

$$\mathrm{FMS}_\tau^{\mathcal{G}}(q) = \mathbb{E}\Big[\sum_{k=1}^{n} \frac{\tilde{e}_k^2 + \alpha_\tau\, \lambda_k^2}{(1 + \alpha_\tau\, \lambda_k)^2}\Big]. \tag{24}$$

At $\tau = 0$ this reads as expected as the squared error of the reported mean. In this paper, we compute the Gaussian form of the score for diagonal covariances $\mathbf{\Sigma}_q(C) = \mathrm{diag}\big(v_1(C), \dots, v_n(C)\big)$, one variance per channel and time step: the eigenbabasis and (24) reduces to a sum over elements of $(e_j^2 + \alpha_\tau\, v_j^2)/(1 + \alpha_\tau\, v_j)^2$.

Since $\mathbf{x}_\tau/\tau = \mathbf{x}_1 + \mathbf{x}_0/\sqrt{\alpha_\tau}$, the noised state carries information on $\mathbf{x}_1$ only in the directions whose variance exceeds $1/\alpha_\tau$. In the eigenbasis of the reported covariance, each mode contributes $(\tilde{e}_k^2 + \alpha_\tau \lambda_k^2)/(1 + \alpha_\tau \lambda_k)^2$ to the Gaussian score: the error of its mean, $\tilde{e}_k^2$, as long as $\alpha_\tau \lambda_k \ll 1$, and a calibration term comparing $\tilde{e}_k^2$ with λ_k once $\alpha_\tau \lambda_k \gg 1$. As τ increases from 0, where only the conditional mean is scored, the uncertainty of the highest-variance modes enters the score first, and that of the lowest-variance modes last. The τ-resolved analysis is therefore a scale analysis ordered by variance: without informative observations the variance follows the energy spectrum, so that, given the spectral decay of geophysical flows, the uncertainty of large scales is evaluated from small τ and that of fine scales only at large τ; with observations, the variance of the large scales is reduced, and the first modes evaluated are those that remain the most uncertain, typically around the effective resolution of the observing system. This reading applies to the joint and sliced versions of the score, not to the per-element one.

We exploit this Gaussian form of the score to assess the calibration of the spread of the ensemble as well as its non-Gaussianity through the two diagnostics detailed below.

τ-resolved spread-calibration diagnostic. We exploit the Gaussian form of the score to assess over- or under-confidence in the spread of a scheme's posterior. Given the context-dependent mean $\mathbf{m}_q(C)$ and covariance $\mathbf{\Sigma}_q(C)$ of q, we define the spread-calibration diagnostic at τ

$$L_\tau = 1 - \frac{\mathrm{FMS}_\tau^{\mathcal{G}}\big(\mathbf{m}_q,\, c^\star(\tau)\,\mathbf{\Sigma}_q\big)}{\mathrm{FMS}_\tau^{\mathcal{G}}\big(\mathbf{m}_q,\, \mathbf{\Sigma}_q\big)}, \qquad c^\star(\tau) = \arg\min_{c>0} \mathrm{FMS}_\tau^{\mathcal{G}}\big(\mathbf{m}_q,\, c\,\mathbf{\Sigma}_q\big). \tag{25}$$

$L_\tau \geq 0$ is the share of the score at τ that a uniform inflation (or deflation) of the reported covariance by the scalar $c^\star(\tau)$ would remove; $c^\star(\tau) > 1$ flags over-confidence and $c^\star(\tau) < 1$ under-confidence. Because the score is computed with the realised errors, L_τ measures the agreement of the reported spread with the scheme's own error, bias included, rather than with the true posterior variance. Since the denominator contains the irreducible term $\mathrm{mmse}_p(\tau)$, L_τ is a lower bound on the corresponding share of the divergence $D_\tau(p \,\|\, q)$.

For small τ, the operator of any q with finite covariance expands as $\Psi_q = \mathbf{m}_q + \tau\, \mathbf{\Sigma}_q\, \mathbf{x}_0 + O(\tau^2)$, so that, with $\mathbf{e} = \mathbf{m}_q - \mathbf{x}_1^\star$,

$$\mathrm{FMS}_\tau(q) = \mathbb{E}\|\mathbf{e}\|^2 + \tau^2\, \mathbb{E}\big[\,\mathrm{tr}(\mathbf{\Sigma}_q^2) - 2\,\mathbf{e}^\top \mathbf{\Sigma}_q\, \mathbf{e}\big] + O(\tau^3). \tag{26}$$

The value at $\tau = 0$ scores the mean component alone; the curvature scores the agreement between the reported covariance and the error of that mean. Minimising the curvature over a scalar inflation $c\,\mathbf{\Sigma}_q$ gives the small-τ limit of (25),

$$c^\star(\tau \to 0) = \frac{\mathbb{E}\big[\mathbf{e}^\top \mathbf{\Sigma}_q\, \mathbf{e}\big]}{\mathbb{E}\big[\,\mathrm{tr}(\mathbf{\Sigma}_q^2)\big]}. \tag{27}$$

For a homogeneous isotropic spread, $\mathbf{\Sigma}_q = v\,\mathbf{I}$ with v the same for every element and context, (27) reduces to $c^\star = \mathbb{E}\|\mathbf{e}\|^2/(n\,v) = 1/\mathrm{SS}^2$, where $\mathrm{SS} = \sqrt{\overline{v}\,/\,\overline{e^2}}$ is the conventional pooled spread–skill ratio [34, 26]. The pooled spread–skill ratio is thus the homogeneous-spread limit of the curvature of the score at $\tau = 0$, and $c^\star(\tau)$ generalises it to every scale. Away from $\tau = 0$ the score weights the elements where the reported spread is small relative to the error, so that $c^\star(\tau)$ departs from $1/\mathrm{SS}^2$ when errors concentrate where the scheme is confident: a scheme can be calibrated in aggregate, $\mathrm{SS} \approx 1$, and still have $L_\tau > 0$ at large τ.

τ-resolved non-Gaussian shape diagnostic. We assess the gain from the non-Gaussian shape of the members, at equal variance, according to

$$G_\tau = 1 - \frac{\mathrm{FMS}_\tau(\mathrm{KDE})}{\mathrm{FMS}_\tau(m, v + h^2)}, \qquad h^2 = \left(1.06\, s\, N^{-1/5}\right)^2, \tag{28}$$

with s the member standard deviation and the denominator the Gaussian score with the same variance as the kernel density. A significantly positive G_τ therefore detects non-Gaussian features of the marginals — skewness, heavy tails, multimodality — that are consistent with the truth; a negative G_τ detects a misleading shape, such as members that are short-tailed relative to the error; and $G_\tau \approx 0$ is inconclusive, the true marginals being close to Gaussian or their non-Gaussianity not being captured by the scheme. By construction $G_0 = 0$, the detection being carried by $\tau > 0$, and the per-element score only sees the marginal distributions.

4 Experimental design

4.1 Three case studies

As experimental testbed, we rely on three dynamical systems of increasing complexity. Each case-study involves forced or coupled dynamics as follows:

- **Lorenz-63 dynamics.** We consider a stochastic Lorenz-63 system [14] augmented with a wind-forcing variable. It is small enough to sweep the model error — bias levels, stated uncertainty, a structural error — and to run weak-constraint 4D-Var and the full ensemble range on every cell. The considered observation configurations account for a regular or random time sampling, the three Lorenz-63 variables being observed.

- **Lorenz-96 dynamics.** We consider the two-scale Lorenz-96 system [36], in the formulation of [24] with a time-scale ratio $\epsilon = 0.1$, with 8 slow and 32 fast variables, augmented with a slow stochastic forcing acting on the slow variables. We assume that only the 8 slow variables and half of the fast ones are observed, meaning that 16 fast variables are never observed.

- **Quasi-geostrophic ocean dynamics.** We consider a two-layer quasi-geostrophic ocean [47] on a 64×64 doubly periodic grid (1000,km, time step 2,h), in the configuration of the pyqg model [1, 42], forced by the wind-stress curl of a low-order chaotic atmosphere built from coupled gyrostats [27], which induces regime changes. Only the upper-layer streamfunction is observed according to randomly-sampled meridional columns mimicking satellite altimetry configurations [25, 3].

For each case-study, we follow the same experimental framework and design two state reconstruction problems, namely a perfect-model experiment (S0) and a imperfect-model one (S1):

- **Perfect-model experiment (S0)**: The model-based schemes assimilate the observations with the same model that generated the truth, including each window's true parameters and stochastic forcing. Their error therefore has only restriction and approximation terms; misspecification is zero by construction;

- **Imperfect-model experiment (S1)**: The truth is unchanged, but the model-based schemes are given an imperfect model: biased parameters, a corrupted forcing and, for L96, a truncated state space (see below). This model error, partly parametric and partly structural, switches on the misspecification term of the chain. The learned schemes are still trained on simulations of the true system, so at S1 they are a reference bound rather than a matched competitor.

We refer the reader to Appendix A for the details of each case study, including its observing systems. For L96, the tables and budget figures use the regular observing system, and the random one is analysed in Sect. 5.5.

For L96, the S1 DA model differs from the true one in three ways:

- **Biased parameters:** F, c_1, h_x, ϵ and the fast coupling weights are all increased by 10 % in every window (h is unchanged);
- **Corrupted forcing:** the DA model receives the true forcing plus a state-dependent term $0.1\,X_1$ and an Ornstein–Uhlenbeck noise (standard deviation 0.71, time scale 5);
- **Truncated state:** the DA model only keeps the 24 observed variables. The 16 unobserved fast variables are removed, so that each slow variable is coupled to its two observed fast variables only, and the fast dynamics run on two variables per slow variable instead of four.

The truncation dominates the resulting model error. Integrated from the true state over one observation interval (100 steps), the S1 DA model has a forecast RMSE of 0.22 on the slow variables and 2.03 on the observed fast ones, against an observation-error standard deviation of 0.71 and climatological standard deviations of 1.91 and 1.66. With the full 40-variable model, the parametric and forcing errors alone give 0.08 and 0.24. S1 thus combines a moderate parametric error with a strong structural error on the fast scales. To separate the two, we also report a variant in which the DA schemes use the full 40-variable model with the same biased parameters and forcing (S1-full). S1 prices model error only for the schemes that consume model information. **[NEEDS RUN: Lorenz-63 only: a sweep of the bias $\delta \in \{0, 5, 10, 20, 40\,\%\}$ crossed with a stated uncertainty $\sigma \in \{0, 20\,\%\}$, plus one structural error.]**

4.2 Model-based DA Schemes

Considered model-based DA baselines. We report numerical experiments with both ensemble Kalman approaches and 4DVar schemes [13]:

- **Ensembles Kalman schemes.** We consider two ensemble filters, namely an Ensemble Transform Kalman Filter (ETKF) [5, 31] and an Enesemble Kalman Filter (EnKF) [21], and one ensemble smoother, the smoother version of the ETKF [5, 31]. As a default configuration, we run all ensemble schemes use 30 members. We also run a 100-member ETKS as a sensitivity row.

- **4D-Var schemes.** we consider both weak-constraint and strong-constraint 4DVar schemes, referred to respectively as Strong-4DVar and Weak-4DVar. They use L-BFGS optimizer and are cycled over sub-windows. The weak-constraint version adds a whitened model-error control at every step, of scale q times the climatological spread [45].

Tuning model-based DA hyper-parameters. All DA hyper-parameters are selected on 50 validation windows, drawn with seeds disjoint from the training and test sets, by sweeping each parameter and retaining the value that minimises the RMSE. For each hyper-parameter, we select a single value per experiment (S0 and S1), shared by the regular and random observing systems. When the two observing systems favour different values, we keep the one with the smallest worst-case RMSE increase relative to the optimum of each observing system:

- Regarding Ensemble Kalman schemes, the multiplicative inflation λ of the analysis anomalies is swept over $\lambda \in [1, 5]$ (13 to 17 values). The selected values are $\lambda = 1.15$ (S0) and 2.5 (S1) for the ETKF, and $\lambda = 1.2$ and 3.0 for the EnKF. These choices cost at most 4.5 % in RMSE relative to the optimum of each observing system. The ETKS uses the inflation of its ETKF filter pass; a separate sweep for the smoother ($\lambda \in [1, 3]$) selects nearly the same values (1.1 and 2.5). For the 100-member ETKS, the inflation is re-selected with the same protocol ($\lambda = 1.05$ and 2.5).
- Regarding 4D-Var schemes, the L-BFGS budget is first increased from 10 to 40 iterations per sub-window, the smallest budget at which the strong-constraint solution converges (validation

RMSE 0.721 at 10 iterations vs 0.545 at 40 at S0). For Weak-4DVar, the model-error scale is then swept over $q \in \{0.003, 0.01, 0.03, 0.1, 0.3, 1\}$ at 40 iterations, giving $q = 0.03$ at S0 and $q = 0.3$ at S1.

These sweeps also characterise how the schemes respond to model error. At S0, the optimal inflation is close to one and the optimal model-error scale is small: Weak-4DVar then performs on par with Strong-4DVar. At S1, both increase by an order of magnitude (λ from about 1.2 up to 2.5–4, q from 0.03 to 0.3). On the regular observing system, the RMSE-optimal ETKF inflation at S1 even yields an over-dispersed ensemble (spread–skill ratio 1.17): in that regime, inflation lowers the RMSE by increasing the weight given to the observations rather than by representing the model error.

4.3 Neural Schemes

Shared backbone. All learned schemes rely on the same neural architecture, a one-dimensional U-Net [41] along the time axis of the assimilation window, built from the diffusion U-Net of the MONAI library [12]. The 24 observed channels of the state are the input channels, and the network processes the full 3000-step window. The encoder has three resolution levels with 64, 128 and 256 channels and two residual blocks per level, with group normalisation (32 groups), dropout 0.1 and no attention layers, for 5.9 M parameters. The observations, with zeros at unobserved times and channels, are concatenated to the input as additional channels, together with the window's parameters $\boldsymbol{\theta}$ and forcing for the schemes that use them. The flow time τ enters every residual block through a sinusoidal embedding; D-Unet uses it with $\tau = 0$ and a zero state input. Using one backbone at a fixed size for all schemes ensures that the differences between learned schemes come from what they learn (a state, a posterior, a prior) and from how they are sampled, not from the network capacity.

Benchmarked neural schemes. We consider the following neural schemes in our experiments:

- **Deterministic UNet (D-Unet).** A deterministic Unet mapping the observations to the state over the window, trained by regression on the true states, with a mean-squared loss plus $0.1\times$ the mean-squared error of the time increments. It estimates the conditional mean and returns no uncertainty.
- **Conditional flow matching (CFM).** CFM learns the posterior $p(\mathbf{x}_1 \mid \mathbf{y})$ along the linear path (11), with source $\mathbf{x}_0 \sim \mathcal{N}(0, 0.5^2 I)$, by regressing the velocity $\mathbf{x}_1 - \mathbf{x}_0$ [35, 2]. The observations enter only as a conditioning input.
- **Score-based DA (SDA).** Following [43], SDA learns a prior on the window states and combines it with an explicit Gaussian likelihood ($\mathbf{R} = 0.5\, I$) at sampling time. The prior is learned by flow matching with the same backbone. We consider three versions of the prior: an unconditional prior (SDA1), a prior conditioned by model parmaters and forcings trained with only true model parameters and forcings (SDA2), the same architecture trained with noisy model parameters and forcings (SDA3). At inference time, SDA comes to leverage the available observations at each integration time step through a correction term of the prior score by the gradient of the observation misfit of the predicted endpoint, scaled by a guidance weight [16].
- **Hybrid (D-Unet+SDA).** The hybrid scheme combines the amortised mean of a D-Unet with the explicit likelihood and prior of an SDA guided sampler. We start the SDA sampling from a noised version of the D-Unet estimate with a given τ value (e.g., $\tau_0 = 0.1$), instead of from pure noise. On QG, the guided sampler started from pure noise does not reach a competitive estimate (validation score 0.32, against 0.85 for D-Unet), so that SDA is only reported in this composed form there.

Training setup. All learned schemes are trained with the same procedure, on simulations of the true system (Sect. 4.1):

- 1000 training and 100 validation windows of 3000 steps, each with its own parameters $\boldsymbol{\theta}$ (20 % spread around nominal) and stochastic forcing;
- a random observing system redrawn with fresh observation noise at every batch: observations at stratified random positions covering the test observing systems;
- per-channel standardisation of the states;
- Adam, learning rate 10^{-3} with cosine annealing, batch size 16, gradient clipping at 10, 1200 epochs;

Inference. CFM draws 30 members per window by integrating the flow ODE with 10 Euler steps from independent draws of $\mathbf{x}_0$. SDA and D-Unet+SDA draw 30 members with 10 guided steps. D-Unet is evaluated with a single forward pass. For schemes truely using model parameters and forcings at inference time, they receive are true model parameters and forcings at `S0` and noisy ones at `S1`, as the model-based schemes do.

Tuning. The training settings are fixed, and only the inference hyper-parameters of the guided schemes are tuned, on 50 validation windows distinct from the test windows, with the same RMSE criterion as for the DA schemes. The SDA guidance weight is set to **[TODO: 20/25]**, which is worth about 8 % in RMSE over the untuned value. For D-Unet+SDA, the start time and the guidance weight are set to $\tau_0 = 0.1$ and 2. D-Unet and CFM have no inference hyper-parameter.

4.4 Evaluation metrics

To compare schemes across the three case studies, whose variables have different units and variability, we report normalised skill scores. Each score S is compared with that of a climatological reference forecast, which issues the climatological distribution of each variable — mean μ_c, variance σ_c^2 — whatever the observations, and is rescaled as

$$\mathrm{Skill}(S) = 1 - \frac{S(q)}{S(\mathrm{clim})}, \tag{29}$$

so that 1 is a perfect estimate, 0 is no better than climatology and negative values are worse than climatology. Skill scores are computed per variable and then averaged:

- L63: the three variables;
- L96: the 24 observed channels (8 slow, 16 fast);
- QG: the two streamfunction layers ψ_1 and ψ_2 ("mean ψ").

Scores are computed at every time step $t = 1, \dots, T$ of the W test windows $w = 1, \dots, W$ ($W = 200$ for L63 and L96, 100 for QG), for every observing system. We denote by $\mathbf{x}_w^\star(t) \in \mathbb{R}^n$ the true state at time t of window w, restricted to the n evaluated variables ($n = 3$ for L63, 24 for L96, the two ψ layers on the 64×64 grid for QG), by $\hat{\mathbf{x}}_w(t)$ the estimate of a scheme — the ensemble mean for sample-based schemes, the analysis trajectory for deterministic ones (Strong- and Weak-4DVar, D-Unet) — and by $\mathbf{x}_w^{(i)}(t)$, $i = 1, \dots, N$, the members of an ensemble. Subscript $j = 1, \dots, n$ denotes a component, e.g. $x_{w,j}^\star(t)$, and $\mu_{c,j}$, $\sigma_{c,j}^2$ are the climatological mean and variance of component j. For QG, the average over the n components is taken within each layer and then over the two layers. Three skill scores are reported: the explained variance (EV) for the mean, the CRPS skill score (CRPSS) for the marginal distributions, and the flow-matching skill score (FMSS_τ) for the posterior at every τ.

Mean component: explained variance. The accuracy of the state estimate is measured by the explained variance, the skill score (29) of the mean-squared error with the climatological

mean as reference,

$$\mathrm{EV} = \frac{1}{n}\sum_{j=1}^{n}\Big(1 - \frac{1}{\sigma^2_{c,j}}\,\frac{1}{WT}\sum_{w=1}^{W}\sum_{t=1}^{T}\big(\hat{x}_{w,j}(t) - x^\star_{w,j}(t)\big)^2\Big). \tag{30}$$

$\mathrm{EV} = \mathrm{FMSS}_0$ (property (i) of Sect. 3). For L96, we also give the per-window RMSE in physical units, $\mathrm{RMSE} = \frac{1}{W}\sum_{w=1}^{W}\frac{1}{n}\sum_{j=1}^{n}\big(\frac{1}{T}\sum_{t=1}^{T}(\hat{x}_{w,j}(t) - x^\star_{w,j}(t))^2\big)^{1/2}$, which the sensitivity analyses of the appendix use.

Spread–skill ratio. For schemes that report an uncertainty, we use the pooled ratio

$$\mathrm{SS} = \left(\frac{\sum_{w,t,j} v_{w,j}(t)}{\sum_{w,t,j}\big(\hat{x}_{w,j}(t) - x^\star_{w,j}(t)\big)^2}\right)^{1/2}, \tag{31}$$

with $v_{w,j}(t)$ the ensemble variance of component j (unbiased estimate over the N members, or the reported analysis variance) [34, 26]. A calibrated N-member ensemble has $\mathrm{SS} = \sqrt{N/(N+1)} \approx 0.98$ for $N = 30$; $\mathrm{SS} < 1$ flags over-confidence. As shown in Sect. 3, SS is the homogeneous-spread, small-τ limit of the spread-calibration diagnostic (25).

Marginal distributions: CRPS skill score. As a marginal check on the uncertainty, we use the continuous ranked probability score of the members for each component [28],

$$\mathrm{CRPS}_{w,j}(t) = \frac{1}{N}\sum_{i=1}^{N}\big|x^{(i)}_{w,j}(t) - x^\star_{w,j}(t)\big| - \frac{1}{2N^2}\sum_{i,k=1}^{N}\big|x^{(i)}_{w,j}(t) - x^{(k)}_{w,j}(t)\big|, \tag{32}$$

and its skill score (29),

$$\mathrm{CRPSS} = \frac{1}{n}\sum_{j=1}^{n}\Big(1 - \frac{1}{\mathrm{CRPS}_{c,j}}\,\frac{1}{WT}\sum_{w=1}^{W}\sum_{t=1}^{T}\mathrm{CRPS}_{w,j}(t)\Big). \tag{33}$$

The reference is the expected CRPS of the climatological distribution of component j, $\mathrm{CRPS}_{c,j} = \frac{1}{2}\,\mathbb{E}|X_j - X'_j|$ with X_j, X'_j independent climatological draws ($\sigma_{c,j}/\sqrt{\pi}$ for a Gaussian climatology), estimated from the test truths. The CRPS is strictly proper for the marginal distributions only. It reduces to the mean absolute error for a deterministic estimate, whose CRPSS is therefore an MAE skill score, reported in parentheses and not ranked against the ensembles. **[NEEDS RUN: CRPS of the ETKF and ETKS from re-runs that store the members; the benchmark files keep only the mean and variance.]**

Posterior: flow-matching skill score. We report FMS_τ (13) at $\tau \in \{0, 0.25, 0.5, 0.75\}$, computed in standardised units (per-variable z-scores with the climatological mean and standard deviation), and its skill score

$$\mathrm{FMSS}_\tau = \frac{1}{n}\sum_{j=1}^{n}\Big(1 - \frac{\mathrm{FMS}_{\tau,j}(q)}{\mathrm{FMS}^{\mathrm{clim}}_{\tau,j}}\Big), \qquad \mathrm{FMS}^{\mathrm{clim}}_{\tau,j} = \frac{\sigma^2_{c,j}}{1 + \alpha_\tau\,\sigma^2_{c,j}}, \tag{34}$$

with $\mathrm{FMS}_{\tau,j}$ the contribution of component j to the score, averaged over windows and times; in standardised units, $\mathrm{FMS}^{\mathrm{clim}}_{\tau,j} = 1/(1+\alpha_\tau)$. The reference is the Gaussian form (24) of the climatological forecast, with mean $\mu_{c,j}$ and variance $\sigma^2_{c,j}$: since $\mathbb{E}(\mu_{c,j} - x^\star_{w,j}(t))^2 = \sigma^2_{c,j}$, it holds whatever the distribution of the truth. $\mathrm{FMSS}_0 = \mathrm{EV}$, so that FMSS_τ extends the explained variance of the mean to the reported uncertainty on the same scale; for $\tau > 0$ its maximum is below 1, set by the irreducible term $\mathrm{mmse}_p(\tau)$. Larger values of τ are dominated by the scheme-independent floor of order $1/\alpha_\tau$, so that $\tau \in [0.25, 0.75]$ is the informative range. Two forms are computed:

- the Gaussian form (24) with diagonal covariance, for every scheme: from the reported mean and variance for the ETKF, EnKF and ETKS, from the member mean and variance for the sample-based schemes, and with $v = 0$ for deterministic schemes, whose score is their mean-squared error at every τ (property (iv));
- the ensemble form (20), with the Silverman bandwidth of (28), for the schemes that output members ($N = 30$).

Unless stated, the score is evaluated per element and pooled over windows, times and channels, with four draws of $\mathbf{x}_0$ per element. As a check of the dependence structure, the sliced version uses 64 random directions per 24-channel snapshot and per space–time patch (24 channels $\times$ 5 times, 25 steps apart, one patch every 100 steps), on times sub-sampled every 10 steps. For a projection $\mathbf{u}^\top \mathbf{x}_1$, the climatological variance in (34) is $\mathbf{u}^\top \boldsymbol{\Sigma}_c \mathbf{u}$, with $\boldsymbol{\Sigma}_c$ the climatological covariance of the patch. The draws of $\mathbf{x}_0$ and of the directions are common random numbers: they are seeded by the test item and shared by all schemes. The degeneracy of the weights in (20) is monitored by the effective sample size $\mathrm{ESS} = 1/\sum_i w_i^2$. For the filter–smoother pair, the restriction identity (18) at $\tau = 0$ is tested by comparing the gap of mean-squared errors with the gap of reported variances, $\frac{1}{WTn}\sum_{w,t,j}\left(v^{q'}_{w,j}(t) - v^{q}_{w,j}(t)\right)$, and with the mean shift $\frac{1}{WTn}\sum_{w,t,j}\left(\hat{x}^{q}_{w,j}(t) - \hat{x}^{q'}_{w,j}(t)\right)^2$, both computed on the same ensemble realisation.

Posterior diagnostics: calibration and shape. Two diagnostics introduced in Sect. 3 complement FMSS_τ for the schemes that report an uncertainty. Both are ratios of a scheme's own scores, computed in standardised units and pooled over windows, times and variables before the ratio is taken. They are therefore dimensionless and comparable across case studies, but they are shares of the scheme's own score, not skill scores with respect to climatology, and we report them in percent.

- **τ-resolved spread–skill ratio.** From the optimal inflation factor $c^\star(\tau)$ of (25), we define $\mathrm{SS}_\tau = 1/\sqrt{c^\star(\tau)}$: 1 for a calibrated scheme, below 1 for an over-confident one and above 1 for an under-confident one. By the small-τ expansion (27), it tends to the pooled spread–skill ratio as $\tau \to 0$ for a homogeneous spread, so that the $\tau = 0$ value of the curve is SS (31). $\mathrm{SS}_\tau = 1$ means that no uniform rescaling of the reported variance improves the score, not that the scheme is calibrated everywhere.
- **Calibration loss.** L_τ (25) is the share of the Gaussian score that the best scalar rescaling $c\,\boldsymbol{\Sigma}_q$ of the reported (diagonal) covariance would remove. A single factor c is used for all windows, times and variables of a scheme, searched over log-spaced values (41 in $[1/8, 8]$ for L96, 49 in $[1/16, 16]$ for QG); the optimum $c^\star(\tau)$ is the inflation ($c^\star > 1$, over-confident) or deflation ($c^\star < 1$) that would calibrate the scheme. When $c^\star$ reaches the edge of the grid, L_τ is a lower bound. $L_0 = 0$ by construction, and the pooled spread–skill ratio is the small-τ limit of $c^\star$ (Sect. 3).
- **Non-Gaussian shape diagnostic.** G_τ (28) compares the ensemble form of the score, with the Silverman kernel density of the members, to the Gaussian form with the same mean and the same variance $v + h^2$. $G_\tau > 0$ when the shape of the members carries information on the truth beyond their mean and variance, $G_\tau < 0$ when it is misleading. It requires members, so it is not available for the schemes that only store a mean and a variance. $G_0 = 0$ by construction.

We report SS_τ and G_τ along τ (Figs. 4 and 5); L_τ quantifies the corresponding loss of score.

Uncertainty of the estimates. Every metric is stored per window. Learned schemes are trained with three seeds; their scores are averaged over seeds per window, and the seed-to-seed range is given separately. We report 95 % intervals from a 2000-replicate window bootstrap. Differences between schemes are paired, by bootstrapping the per-window differences, since all schemes are scored on the same windows, observations and draws. The DA ensembles are not seeded: two realisations of the L96 ETKS differ by about 2 % in RMSE at `S0`, a realisation noise

Table 2: **Perfect-model experiment (S0).** Explained variance (EV) and CRPS skill score (CRPSS) against climatology (Sect. 4.4; 1 is perfect, 0 is climatology). L96: 24 observed channels, regular observing system, 200 test windows; learned rows averaged over 3 seeds. QG: mean over ψ_1, ψ_2, 3 altimetry columns per day, 100 test windows; DA ensembles with $N = 80$ members, learned D-Unet (G1-L) and D-Unet+SDA (SDA warm-started from G1-L). SS: pooled spread–skill ratio (31) (about 0.98 for a calibrated 30-member ensemble; below 1, over-confident). Values in parentheses are MAE skill scores of deterministic schemes, not ranked against the ensembles; † estimated from the RMSE assuming Gaussian errors (a lower bound by up to 0.04). Empty cells: not available. SDA denotes score-based DA with a prior conditioned on the model parameters $\boldsymbol{\theta}$ and the forcing, trained with the true parameters and forcing (SDA2 in Sect. 4.3). For L96, D-Unet+SDA uses the SDA3 prior, trained with noisy parameters and forcing. Best per column in bold, second best in italics.

	L63			L96			QG		
scheme	EV	CRPSS	SS	EV	CRPSS	SS	EV	CRPSS	SS
ETKF				0.815	0.710	0.76	0.934	0.867	
EnKF				0.801	0.693	0.85	0.923	0.857	
ETKS				0.866	0.744	0.48	0.963	*0.898*	
Strong-4DVar			–	0.741	(0.523)	–	**0.980**	(0.885)†	–
Weak-4DVar			–	0.757	(0.520)	–			–
D-Unet			–	*0.950*	(0.774)	–	0.975	(0.888)	–
CFM				0.942	*0.838*	0.93			
SDA				0.910	0.772	0.81			
D-Unet+SDA				**0.961**	**0.853**	0.56	*0.977*	**0.912**	

that the window bootstrap does not include.

5 Results

This section reports the experiments conducted on the three case studies, namely Lorenz-63, Lorenz-96 and QG ocean dynamics, introduced in Section 4.1. Section 5.1 gives the error budget with a perfect model, and Sect. 5.2 its change under model error. Section 5.3 focuses on uncertainty quantification with the flow-matching score introduced in Section 3. We further analyse the sensitivity to the structure of the prior and to the observing system in Sections 5.4 and 5.5.

5.1 Error budget with a perfect model

Overall performance. Table 2 reports the performance of all schemes with a perfect model. On L96, the learned schemes outperform model-based DA. D-Unet+SDA is best for both the mean and the distribution (EV 0.961, CRPSS 0.853), followed by D-Unet (EV 0.950) and CFM (EV 0.942, CRPSS 0.838). The best DA scheme, the ETKS (EV 0.866, CRPSS 0.744), is behind every learned scheme, including SDA (EV 0.910, CRPSS 0.772), and the 4D-Var schemes rank last (EV 0.74–0.76), behind the ensemble filters (0.80–0.82). These differences come almost entirely from the fast variables. On the 8 slow variables, all schemes reach an EV of 0.96–0.99, and the ETKS (0.988) is within 0.01 of the best scheme (0.995). On the 16 observed fast variables, the DA schemes reach 0.63–0.80, against 0.87–0.94 for the learned schemes. On QG, the schemes are much closer. Strong-4DVar has the best mean (EV 0.980), on par with D-Unet+SDA (0.977) and D-Unet (0.975), and the ETKS remains the best DA ensemble (EV 0.963). For the distribution, D-Unet+SDA again leads (CRPSS 0.912), ahead of the ETKS (0.898) and the filters (0.86–0.87). In both case studies, smoothing improves on filtering, and the ETKF slightly outperforms the EnKF. The spread–skill ratios qualify these rankings on L96:

CFM is nearly calibrated (SS 0.93), whereas the DA ensembles are over-confident, the more so as they are more accurate (EnKF 0.85, ETKF 0.76, ETKS 0.48). SDA is moderately over-confident (0.81), and D-Unet+SDA, despite the best mean and CRPSS, markedly so (0.56), a consequence of its tempered guidance (Sect. 5.4). **[TODO: QG spread–skill ratios.]** The following paragraphs analyse where the gaps to the best scheme come from.

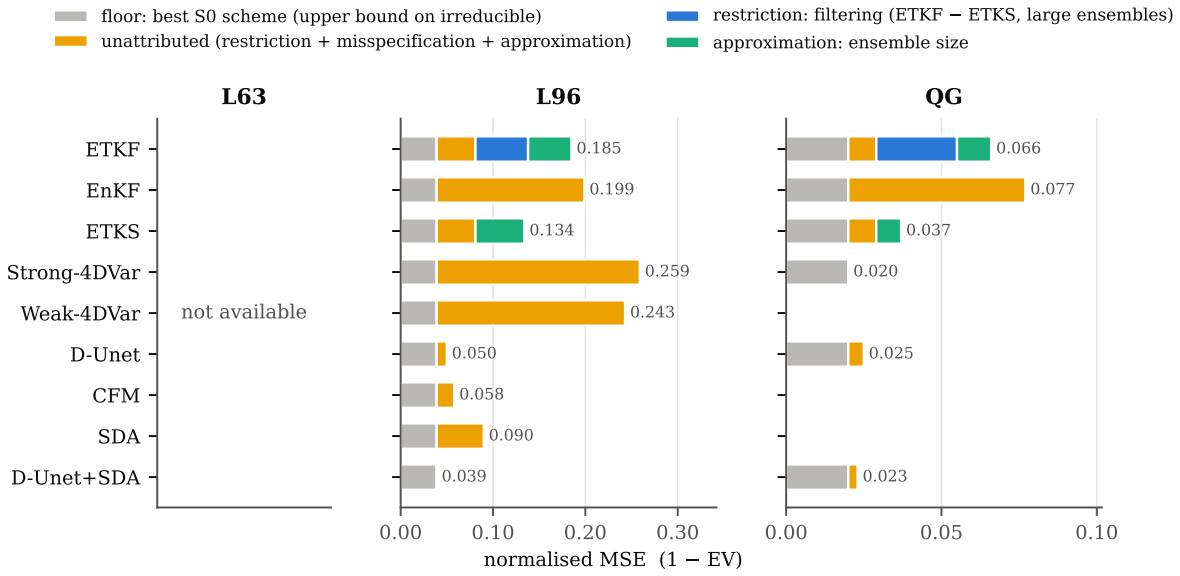


Figure 1: **Error budget in the perfect-model experiment (S0).** Normalised mean-squared error ($1-\mathrm{EV}$, Table 2) of each scheme, split along the error chain (10). Grey: the best S0 scheme of each case study, an upper bound on the irreducible term. Blue: restriction due to filtering, measured as the gap between the ETKF and the ETKS with large ensembles (100 members for L96, 320 for QG). Green: approximation due to the ensemble size (30 vs 100 members for L96, 80 vs 320 for QG). Orange: excess over the floor not attributed to a term by a contrast; it can contain restriction, approximation and, through the posterior family of the scheme (e.g. Gaussian updates, mode instead of mean), misspecification, even with a perfect dynamical model.

Irreducible component. The best scheme sets the floor of each case study: D-Unet+SDA on L96 (normalised mean-squared error $1-\mathrm{EV}=0.039$) and Strong-4DVar on QG (0.020). It is only an upper bound on the irreducible term mmse in (10): we have no lower bound, so every component below is a share of the excess over this floor, not over the true irreducible error.

Restriction component in filtering schemes. A filter conditions each state on past observations only, a smoother on the whole window. Measured with large ensembles, so that the ensemble size does not contaminate it, filtering costs 0.057 of normalised MSE on L96 (ETKF 0.138 against ETKS 0.081, 100 members) and 0.026 on QG (0.055 against 0.029, 320 members). It is the largest identified term of the DA ensembles in both case studies. To check that this gain is a restriction, we run the filter and the smoother on the same L96 ensemble realisation and evaluate both sides of the identity $\mathrm{MSE}_F-\mathrm{MSE}_S=\mathrm{var}_F-\mathrm{var}_S=\mathbb{E}\|m_F-m_S\|^2$ (physical units). It holds approximately with 100 members — an MSE gap of 0.162 against a variance gap of 0.176 and a mean shift of 0.233 — and less well with 30 (0.159, 0.227, 0.296). The deviation shrinks as the ensemble grows, as expected if it is an approximation error of the ensemble.

Approximation components. We can partly assess the approximation components of the different schemes:

- **Ensemble size in ensemble DA approaches.** Going from 30 to 100 members reduces the normalised MSE of the L96 ETKS from 0.134 to 0.081 (40 %) and that of the ETKF from 0.185 to 0.138. On QG, going from 80 to 320 members (with a re-tuned localisation radius) reduces it from 0.037 to 0.029 for the ETKS and from 0.066 to 0.055 for the ETKF.
- **Optimisation in 4DVar schemes.** With 10 L-BFGS iterations per sub-window, Strong-4DVar is far from its own optimum (validation RMSE 0.721 against 0.545 with 40 iterations). The benchmark uses 40 iterations, at which the solution has converged, so the remaining excess of 4D-Var is not an optimisation error.
- **Training setup and architecture design in neural schemes.** Training D-Unet for 1200 instead of 400 epochs reduces its L96 RMSE from 0.380 to 0.340, and three times more training windows only to 0.334: the approximation term of the learned schemes is close to saturation at our budget. Complementary experiments with three tiers of the shared U-Net backbone, under an earlier protocol (fixed observing system, 400 epochs, one seed; Appendix F), support that network capacity is not the limiting approximation term: a larger tier (23.5 M parameters) brings no gain, while a smaller one with four times fewer parameters (1.5 M) increases the RMSE by 4 to 24 % (e.g. 0.490 vs 0.470 for D-Unet and 0.410 vs 0.341 for CFM on L96).

Unattributed component. The excess left after these contrasts is therefore unattributed: it can contain restriction, approximation and structural misspecification. On L96, it is 0.042 for the 100-member ETKS, and it makes up almost all of the excess of the 4D-Var schemes (0.20 to 0.22), although they use the exact model and have converged. Their candidates are structural: the mode instead of the mean, a hard constraint, a Gaussian background, and the restriction of cycling over sub-windows. The EnKF exceeds the ETKF (0.199 against 0.185), consistent with the sampling noise of its perturbed observations. For the neural schemes, the unattributed term is smaller: 0.011 for D-Unet, 0.019 for CFM and 0.051 for SDA, whose guided sampler approximates the likelihood. On QG, the same holds for the ensembles and the neural schemes, at a smaller scale: the unattributed term is 0.009 for the 320-member ETKS, the EnKF again exceeds the ETKF (0.077 against 0.066), and the neural schemes lie within 0.005 of the floor. The variational scheme is the exception: Strong-4DVar sets the QG floor itself, so the structural costs that dominate its L96 budget do not appear on the streamfunction. They are not absent, however: on the potential vorticity, which is smaller-scale and unobserved, D-Unet and D-Unet+SDA outperform it (EV 0.743 and 0.751 against 0.700).

Synthesis. Figure 1 summarises the budget per scheme family. For the ensemble DA schemes, the excess over the floor splits into comparable shares of restriction, approximation and unattributed error on L96 (ETKF: 0.057, 0.047 and 0.042), while the restriction due to filtering dominates on QG (ETKF: 0.026, 0.011 and 0.009). For the variational schemes, the excess is almost entirely unattributed on L96, while Strong-4DVar is the best scheme on QG. By contrast, the end-to-end neural schemes, D-Unet and CFM, lie just above the floor in both case studies (excess 0.011 and 0.019 on L96, 0.005 for D-Unet on QG). SDA, which combines a learned prior with an explicit likelihood, has a larger excess on L96 (0.051), in the same range as the unattributed term of the 100-member ETKS (0.042). This points to a misspecification of the posterior induced by the approximate likelihood of the guided sampler, even though the observation operator and the error statistics are exact. On QG, SDA alone does not reach a competitive estimate; started from the D-Unet estimate, it stays within 0.003 of the floor (D-Unet+SDA). Overall, even with a perfect dynamical model, the error budget of the DA schemes involves significant approximation and misspecification components, beyond the restriction due to their conditioning. These components prevent the schemes from extracting the full information that the observations carry about the state. As a consequence, these findings suggest that the skill of end-to-end neural schemes deliver a better and more robust indicator of the reconstruction skill that a given observing system actually allows: assessments of observing systems based on DA alone, such as observing

Table 3: **Imperfect-model experiment (S1).** Same metrics, test sets and conventions as Table 2. The DA hyper-parameters are tuned for S1 (Sect. 4): inflation and model-error scale for L96; for QG, the observation-error variance multiplied by 6 for the ensembles and the weak constraint for Weak-4DVar, while Strong-4DVar keeps its S0 setting. The learned schemes are trained on simulations of the true system: at S1 they are an oracle bound, not a matched competitor. SDA and D-Unet+SDA receive the biased parameters and forcing given to the model-based schemes. For L96, the S1 DA model is also truncated to the 24 observed variables (Sect. 4.1). † estimated from the RMSE assuming Gaussian errors (a lower bound by up to 0.04). Best per column in bold, second best in italics.

	L63			L96			QG		
scheme	EV	CRPSS	SS	EV	CRPSS	SS	EV	CRPSS	SS
ETKF				0.272	0.201	0.59	0.828	0.787	
EnKF				0.266	0.196	0.72	0.829	0.783	
ETKS				0.358	0.191	0.34	0.838	*0.819*	
Strong-4DVar			–	0.239	(−0.042)	–	0.779	(0.709)†	–
Weak-4DVar			–	0.561	(0.225)	–	0.847	(0.764)†	–
D-Unet			–	*0.950*	(0.774)	–	*0.964*	(0.846)	–
CFM				0.943	*0.838*	0.94			
SDA				0.905	0.766	0.95			
D-Unet+SDA				**0.961**	**0.854**	0.58	**0.965**	**0.871**	

system simulation experiments, may underestimate the information content of the observations.

5.2 Error budget under model error

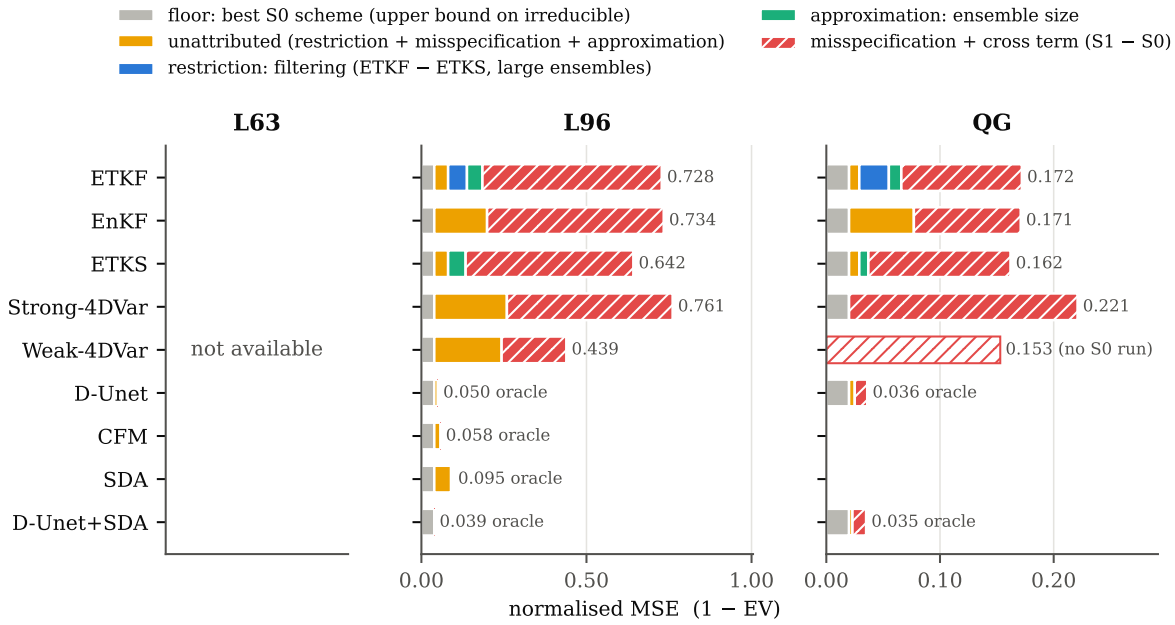


Figure 2: **Error budget under model error (S1).** The S0 budget of Fig. 1, extended by the increase of the normalised mean-squared error from S0 to S1 (hatched red), i.e. misspecification plus the cross term of (10) (Sect. 2). Learned schemes are trained on the true system and are an oracle bound at S1. For L96, the S1 DA model is also truncated to the 24 observed variables (Sect. 4.1). QG Weak-4DVar is run at S1 only (outline).

Overall performance. Table 3 reports the performance under model error. On L96, model-based DA degrades strongly: the EV of the ETKF, EnKF, ETKS and Strong-4DVar falls to 0.24–0.36, and the CRPSS of the DA ensembles to about 0.2. Weak-4DVar is the best DA scheme (EV 0.561). The degradation is again much larger on the fast variables: the slow variables keep an EV of 0.61–0.67 (0.86 for Weak-4DVar), while the fast ones fall to 0.03–0.23 (0.41 for Weak-4DVar). The DA ensembles also become more over-confident: their spread–skill ratio drops to 0.59 for the ETKF, 0.72 for the EnKF and 0.34 for the ETKS, while the learned schemes keep their S0 values (CFM 0.94, SDA 0.95, D-Unet+SDA 0.58). **[TODO: QG spread–skill ratios.]** On QG, the DA schemes keep an EV of 0.78–0.85, with Weak-4DVar best for the mean (0.847) and the ETKS best for the distribution (CRPSS 0.819). The learned schemes are almost unaffected (L96 EV 0.905–0.961, QG 0.964–0.965), with D-Unet+SDA again best (CRPSS 0.854 on L96, 0.871 on QG). For L63 and L96, the S1 test data remain within the training distribution of the neural schemes, since only the model given to DA changes. For QG, the S1 observation errors (15 % white and 15 % correlated per pass) lie at the upper edge of the noise range used to train D-Unet (5–15 % white, 0–15 % correlated), while the SDA prior does not see the observations, which enter only through its likelihood. The QG experiments thus also test the robustness of the neural schemes to observation errors at the limit of their training distribution.

Impacts of misspecification. The increase of the normalised MSE from S0 to S1 measures misspecification plus the cross term of the chain (Fig. 2, hatched red), and its effect differs between scheme families. For the ensemble DA schemes, the increase is 0.50–0.54 on L96, i.e. about three times the S0 error of the ETKF and EnKF and nearly four times that of the ETKS, and 0.09–0.13 on QG. The contrasts that reduced their S0 excess no longer help: 100 members leave the L96 ETKS unchanged (EV 0.358 with 30 and 100 members), 320 members barely improve the QG ETKS (0.838 to 0.844), and smoothing gains only 12 % of normalised MSE over filtering on L96 (0.642 against 0.728) and 6 % on QG (0.162 against 0.172). The filter/smoother identity also fails: on L96, the MSE gap is 0.237, against a variance gap of 0.534 and a mean shift of 0.913. Re-tuning the inflation (L96) or the observation-error variance (QG) for S1 recovers only part of the loss, e.g. from EV 0.809 to 0.838 for the QG ETKS. For the variational schemes, the hard constraint is the least robust: the increase is 0.50 on L96 and 0.20 on QG, where Strong-4DVar goes from the best scheme with a perfect model to the worst DA scheme under model error. A weak constraint, with its model-error scale tuned per case, gains little at S0 (EV 0.757 against 0.741 on L96), as expected, but reduces the normalised MSE of Strong-4DVar by 42 % on L96 (0.439 against 0.761) and 31 % on QG (0.153 against 0.221) at S1, making Weak-4DVar the best DA scheme for the mean. It remains, however, about ten times the learned bound on L96: an additive per-step model error with a scalar scale cannot absorb a parametric bias. By contrast, the neural schemes are barely affected. D-Unet, CFM and D-Unet+SDA are flat on L96, and SDA loses only 0.005, as its prior, trained with the true parameters, receives the biased ones (Sect. 5.4). On QG, they lose about 0.01 of normalised MSE, the effect of the S1 observation errors, the only part of S1 they see.

Synthesis. Under model error, misspecification dominates the budget of every model-based scheme, and the restriction and approximation terms of S0 become secondary (Fig. 2). The observations still bring information to the model-based schemes: on QG, a free run of the biased model has a negative EV on the streamfunction (−0.40), against 0.78–0.85 after assimilation. But misspecification caps what they can extract, and the explicit representations of model uncertainty available to DA (a weak constraint, inflation, an inflated observation error) recover only part of it. The neural schemes, which do not run a dynamical model at inference time, are insensitive to the misspecification of the model given to DA: on L96 they reach an EV above 0.9 from the same observations, against at most 0.56 for DA. This gap is, however, an upper bound on what the observations allow under model error: the neural schemes are trained on simulations of the

true system, so the gap also contains the value of that training data. **[NEEDS RUN: Neural schemes trained on simulations of the biased model, with $\sigma = 0$ and 20 %; their gap to the rows trained on the true system measures what training on reanalyses or true-system data implicitly brings.]** Under model error, the skill of a DA scheme therefore reflects the fidelity of its model as much as the information content of the observations, which further limits its use to assess an observing system.

5.3 τ-resolved evaluation of the reported uncertainty

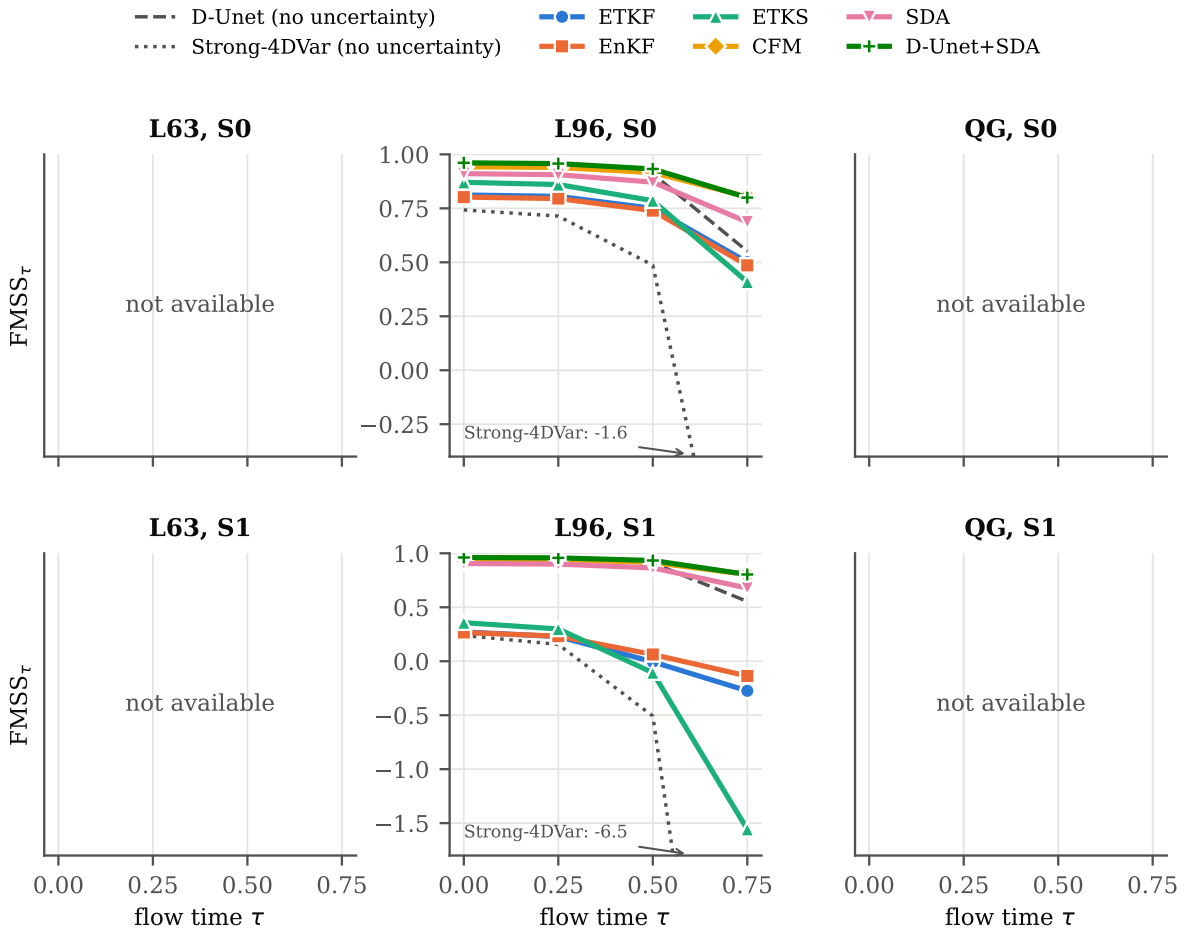


Figure 3: **Skill of the reported posterior along the flow time.** Flow-matching skill score FMSS_τ (34) (Gaussian form) at $\tau \in \{0, 0.25, 0.5, 0.75\}$, perfect-model (top) and imperfect-model (bottom) experiments; FMSS_0 is the explained variance. Deterministic schemes (grey) score their mean-squared error at every τ. Learned rows averaged over 3 seeds. **[TODO: L63 and QG panels.]**

Skill of the reported posterior. Figure 3 extends the explained variance of Tables 2 and 3 ($\tau = 0$) to the reported distribution ($\tau > 0$). With a perfect model, all schemes lose skill as τ grows, but at different rates. The DA ensembles fall from 0.80–0.87 to 0.41–0.50 at $\tau = 0.75$, and their ranking reverses: the ETKS, best for the mean, still leads at $\tau = 0.5$ (0.79 against 0.75 for the ETKF) but ends below both filters (0.41 against 0.50 and 0.49). CFM keeps most of its skill (0.94 to 0.80) and reaches the level of D-Unet+SDA (0.80) at $\tau = 0.75$, although its mean is less accurate. Reporting no uncertainty is penalised: D-Unet, as accurate as the best

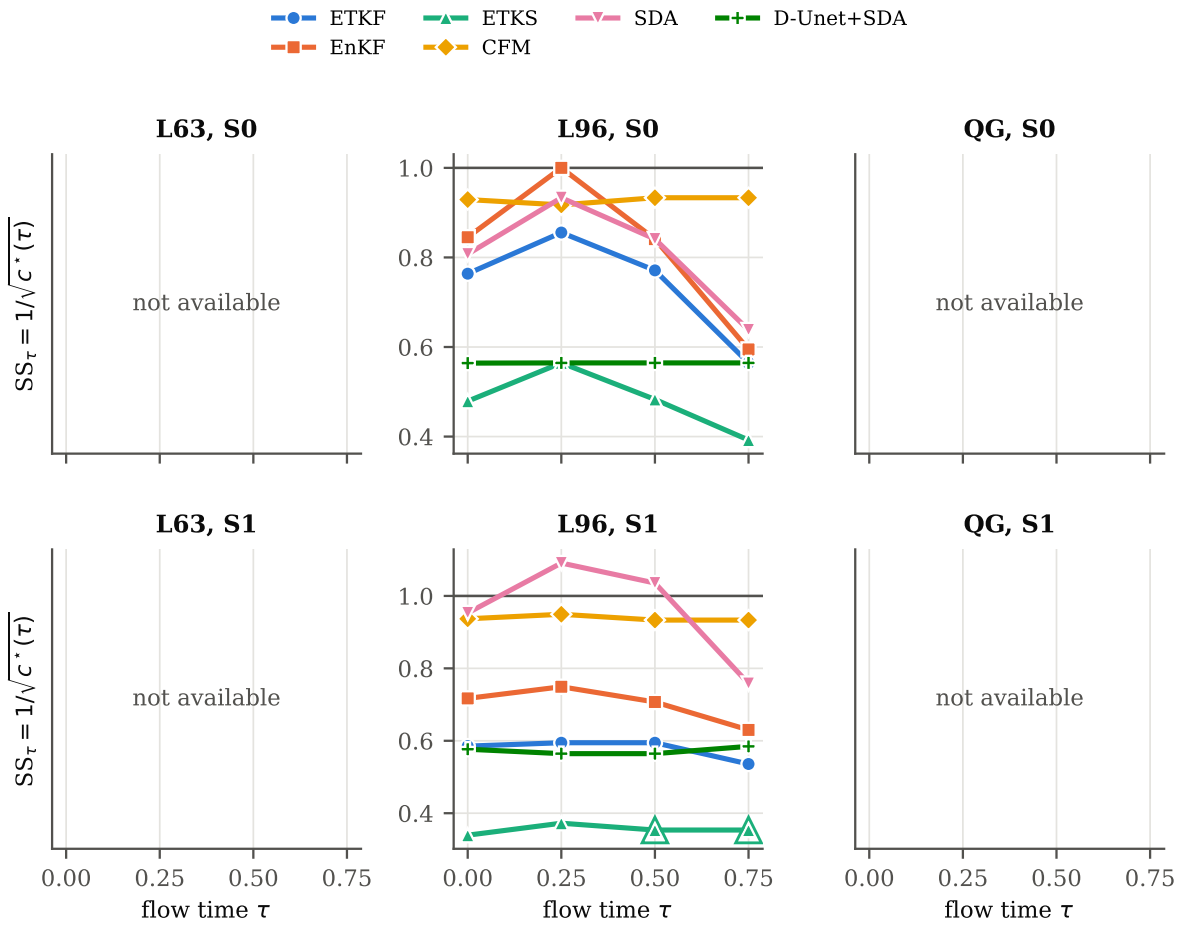


Figure 4: **Calibration of the reported spread along the flow time.** τ-resolved spread–skill ratio $\mathrm{SS}_\tau = 1/\sqrt{c^\star(\tau)}$ (1: calibrated; below 1: over-confident); its value at $\tau = 0$ is the pooled spread–skill ratio. Open markers: optimal inflation at the edge of the search grid, so that SS_τ is an upper bound. **[TODO: L63 and QG panels.]**

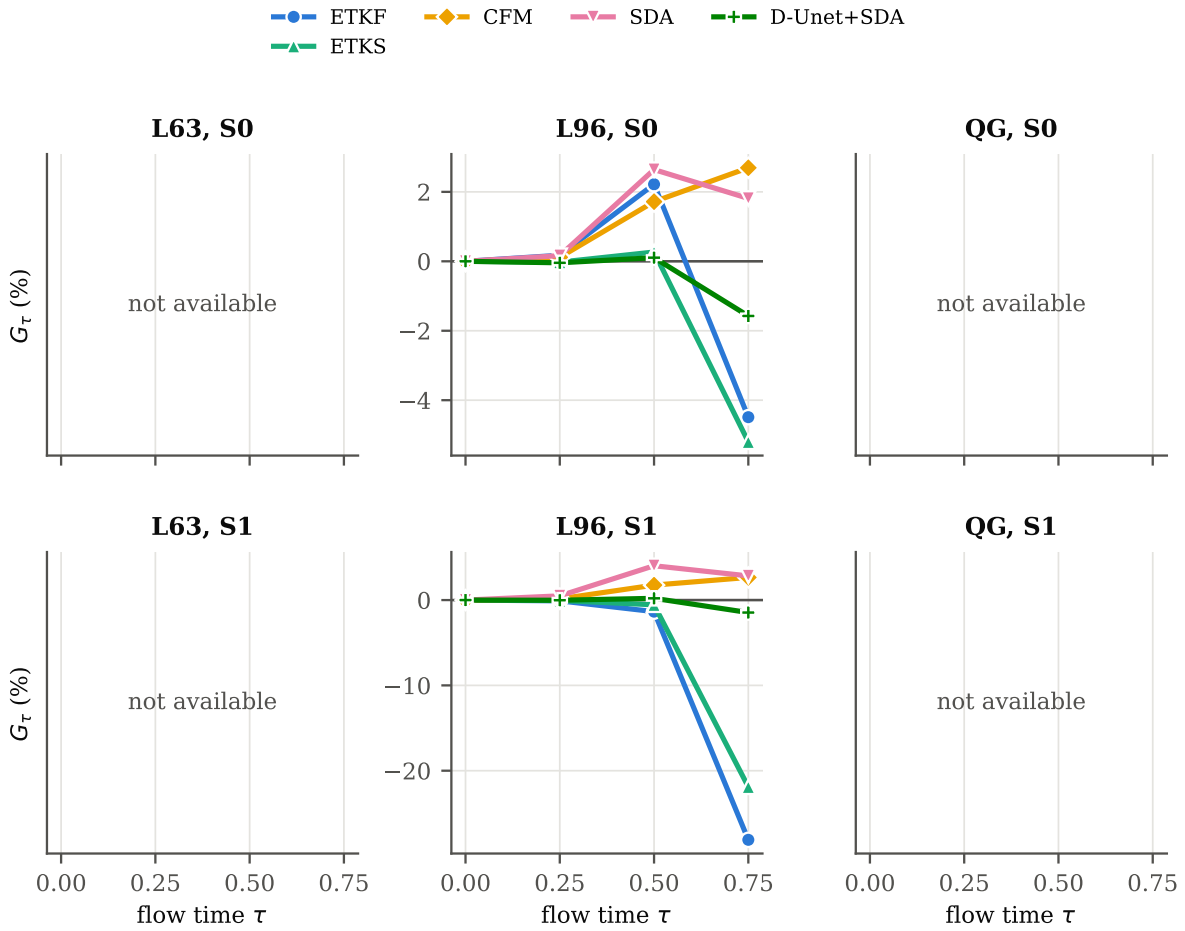


Figure 5: **Information carried by the shape of the reported distribution.** Non-Gaussian shape diagnostic G_τ (28), in percent (0: no gain over a Gaussian of the same variance; positive: informative shape; negative: misleading shape). It requires members, so the EnKF is not shown. **[TODO: L63 and QG panels.]**

learned schemes for the mean, falls to 0.55, and Strong-4DVar to -1.6. Under model error, the decay of the DA schemes becomes much steeper: their FMSS crosses zero, i.e. the climatological forecast, as soon as $\tau \approx 0.5$, and reaches -1.56 for the ETKS at $\tau = 0.75$. The learned schemes keep their S0 curves. Misspecification thus degrades the reported distribution even more than the mean. **[TODO: QG and L63.]**

Calibration and shape. The τ-resolved spread–skill ratio (Fig. 4) and the shape diagnostic (Fig. 5) explain these differences. CFM is calibrated at every τ and in both experiments (SS_τ 0.92–0.95). The DA ensembles are over-confident, and the more so the more accurate their mean: at S0, $\mathrm{SS}_{0.75}$ is 0.56 for the ETKF, 0.59 for the EnKF and 0.39 for the ETKS, and at S1 it drops to 0.54, 0.63 and at most 0.35. The smoother thus buys its better mean with an over-confident ensemble, and model error worsens it. D-Unet+SDA is uniformly over-confident ($\mathrm{SS}_\tau \approx 0.56$), the cost of its tempered guidance, while SDA is close to calibrated at small τ and over-confident at large τ (0.64 at S0). A calibrated pooled spread does not imply a calibrated distribution: on the random observing system at S1, the ETKF has a pooled spread–skill ratio of 0.99, yet a scalar rescaling still removes 12 % of its score at $\tau = 0.75$, as its spread does not match its error where the error occurs. Regarding the shape of the distribution, the diagnostic has a small negative floor: on synthetic Gaussian posteriors, 30 members drawn from the true distribution give $G_{0.75}$ between 0 and -1.4 %, and over-confident Gaussian members down to -4 %, with no shape effect at all. Within this floor, the shape of the DA ensembles at S0 (-4.5 % for the ETKF, -5.2 % for the ETKS) cannot be distinguished from the effect of their over-confidence. Under model error, by contrast, their shape is clearly misleading (-22 % for the ETKS, -28 % for the ETKF): their members are short-tailed with respect to their errors, so that the truth falls outside the ensemble more often than its spread allows. The nearly calibrated neural ensembles show a small but genuine gain ($G_{0.75}$ of $+2$ to $+3$ % for CFM and SDA): their shape carries some information beyond their mean and variance. **[TODO: QG and L63.]**

Synthesis. The τ-resolved scores separate what the mean-based metrics of Sects. 5.1 and 5.2 conflate. With a perfect model, the DA ensembles lose more skill from the mean to the distribution than the learned schemes, because they are over-confident, and the more so as their mean improves; smoothing and larger ensembles reduce the error of the mean, not the miscalibration. Under model error, misspecification affects the reported distribution even more than the mean: the DA ensembles become more over-confident and, beyond the noise floor of the shape diagnostic, short-tailed, and their score at large τ falls below that of climatology. The learned schemes, by contrast, report a distribution whose skill, calibration and shape are unchanged from S0 to S1, with CFM the best calibrated. Their advantage over DA is therefore larger for the distribution than for the mean. These conclusions hold when the score is evaluated on random projections of space–time patches rather than per element (Appendix G), which accounts for the dependence between variables. **[NEEDS RUN: The filter/smoother identity along τ (score of the ETKS's own filter pass).]**

5.4 Sensitivity to the structure of the prior

The posterior (3) is a likelihood times a prior. Model-based DA derives the prior from the equations, whereas SDA learns it from data and keeps the same explicit likelihood. Comparing them isolates the role of the prior. Table 4 gathers the L96 schemes that share this likelihood, grouped by the origin of their prior.

Same likelihood, two priors. With a perfect model, the learned joint-window prior (SDA, EV 0.908–0.911) outperforms the model-derived priors of the hard constraint (Strong-4DVar, 0.741) and of the 30-member smoother (ETKS, 0.866), although these use the exact equations. It

Table 4: **Prior structure at matched likelihood (L96).** Explained variance (EV), flow-matching skill score at $\tau = 0.75$ and pooled spread–skill ratio (SS) in the perfect-model (S0) and imperfect-model (S1) experiments, regular observing system. All schemes use the same explicit observation operator and error covariance. Learned rows averaged over 3 seeds. The italic row is a sensitivity row.

		EV		$\text{FMSS}_{0.75}$		SS	
prior	scheme	S0	S1	S0	S1	S0	S1
model, hard constraint	Strong-4DVar	0.741	0.239	−1.56	−6.48	–	–
model + model error	Weak-4DVar	0.757	0.561	−1.40	−3.32	–	–
model, 30-member ensemble	ETKS	0.866	0.358	0.41	−1.56	0.48	0.34
model, 100-member ensemble	*ETKS*	*0.919*	*0.358*				
learned, unconditional	SDA1	0.908	0.908	0.62	0.62	0.61	0.61
learned, $\boldsymbol{\theta}$ + forcing (true training)	SDA2	0.910	0.905	0.69	0.68	0.81	0.95
learned, $\boldsymbol{\theta}$ + forcing (noisy training)	SDA3	0.911	0.911	0.66	0.66	0.67	0.69
learned + amortised mean	D-Unet+SDA	0.961	0.961	0.80	0.80	0.56	0.58

does not outperform the 100-member smoother (0.919): the ordering is set by the approximation term of the model-based scheme, not by the fidelity of its prior, which is exact at S0. The learned prior also gives the better reported distribution ($\text{FMSS}_{0.75}$ of 0.62–0.69 against 0.41 for the ETKS), as it is less over-confident (SS 0.61–0.81 against 0.48).

A better prior is not a better estimate. Two observations make the point. Training the SDA prior three times longer (400 to 1200 epochs) improves the estimate by 7 to 9 % in RMSE, and the true equations, used as a hard prior, give the worst DA scheme at S0. What matters is how much of the prior the scheme can actually use within its approximation budget. This reading also resolves the controlled result that motivated us [22]: a learned prior can beat the true model because of the approximation and restriction terms of the scheme that uses the true model, not because the learned prior is better.

Conditioning the prior. Conditioning the learned prior on the window parameters $\boldsymbol{\theta}$ and forcing brings little with a perfect model: SDA1, which ignores them, reaches an EV of 0.908, against 0.910 and 0.911 for SDA2 and SDA3, i.e. 2–3 % of normalised MSE. Under model error, this conditioning becomes a misspecification channel when the prior was trained with exact parameters: SDA2, given the biased parameters, loses 0.005 of EV and leaks the bias into the D-Unet+SDA scheme built on it (RMSE 0.310 to 0.333 with 400-epoch priors), whereas SDA3, trained with noisy parameters, is unaffected. **[NEEDS RUN: Value of the conditioning information for DA and SDA: the same test windows with nominal parameters and/or no forcing given to the schemes (ETKF, EnKF, ETKS, Strong-4DVar with the full model; SDA2, SDA3).]**

Composing a mean and an uncertainty. In the operator language of Sect. 3, D-Unet+SDA composes D-Unet's mean with the anomaly of SDA3's guided sampler. It has the best mean in both experiments (EV 0.961), but not the best distribution: CFM reaches the same $\text{FMSS}_{0.75}$ (0.80) with a less accurate mean, because D-Unet+SDA is over-confident (SS 0.56), a consequence of the tempered guidance of its sampler. The score thus separates the two components D-Unet+SDA composes: an amortised mean, accurate, and an explicit-likelihood sampler whose uncertainty is too narrow. The composition matters even more on QG: there, the learned prior alone, guided by the observations from pure noise, does not reach a competitive estimate, whereas started from the D-Unet estimate it gives the best QG scheme for both the mean and the

distribution (Table 2). **[TODO: L63.]**

5.5 Observing system and marginal value of observations

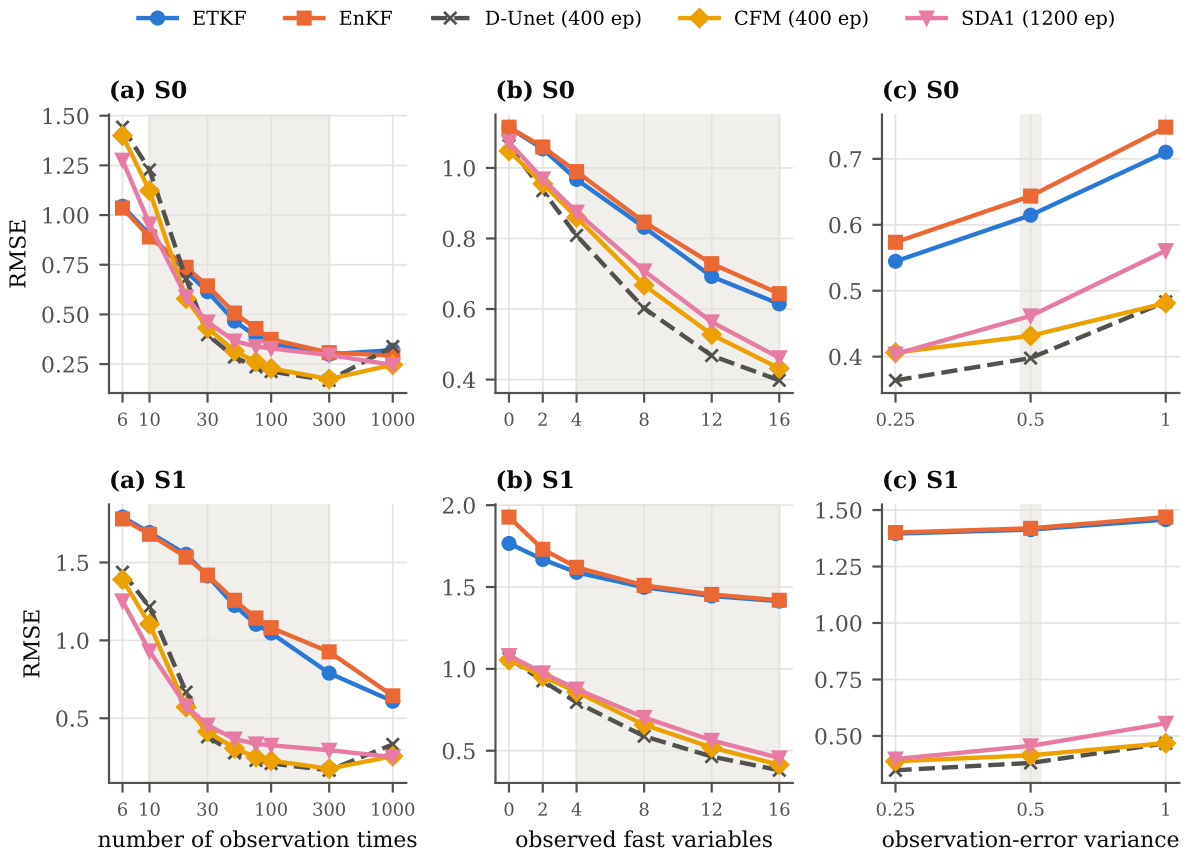


Figure 6: **Response to the observing system, inside and outside the training range (L96).** RMSE on the observed variables against (a) the number of observation times per window (16 observed fast variables), (b) the number of observed fast variables (30 observation times) and (c) the observation-error variance (30 times, 16 fast variables), in the perfect-model (top) and imperfect-model (bottom) experiments; 20 windows × 3 draws of the observing system per point. Shaded: range seen by the neural schemes during training. **[TODO: Preliminary version from existing runs: D-Unet and CFM at 400 epochs, SDA1 at 1200 epochs, seed 1; ETKF and EnKF only. To be replaced by the benchmark schemes (1200 epochs, SDA, D-Unet+SDA, ETKS, Strong-4DVar) and by EV.]**

Generalisation beyond the training observing systems. Figure 6 shows how the schemes respond to the observing system. Within the range seen in training, the neural schemes follow the same trends as DA and outperform it as soon as a window holds about twenty observation times; with ten or fewer, DA is better (RMSE 0.90 for the ETKF against 0.96–1.23 for the neural schemes at ten times, S0). Outside this range, their behaviour depends on how they use the observations. With fewer observations than in training (6 times), all neural schemes are worse than DA (1.28 to 1.44 against 1.04). With more (1000 times), D-Unet degrades (0.17 at 300 times, 0.34 at 1000) and CFM to a lesser extent (0.17 to 0.25), whereas SDA, whose likelihood is explicit, keeps improving (0.30 to 0.25), like DA. With fewer observed fast variables than in training (0 or 2), all schemes degrade smoothly and the neural ones keep their advantage. When the observation-error variance departs from its training value, all schemes respond in the

expected direction, including D-Unet, which is not told the variance (0.36 at 0.25, 0.40 at 0.5, 0.48 at 1.0). Across observing systems, keeping the observation operator and error covariance explicit also buys robustness: the ratio of random to regular RMSE is 1.06 for Strong-4DVar, 1.15 for the ETKS and 1.21–1.24 for SDA, against 1.32 for CFM and D-Unet. In the language of the chain, an amortised map becomes misspecified once the observing system leaves its training distribution, whereas a scheme with an explicit likelihood does not.

Marginal value of observations. The slope of the curves of Fig. 6a measures what additional observations bring to each scheme. With a perfect model, going from 10 to 100 observation times divides the RMSE by 2.6 for the ETKF (0.90 to 0.35) and by 3 to 6 for the neural schemes. Under model error, the neural schemes keep the same curves, whereas the DA schemes start from a much higher error and gain far less: the ETKF goes from 1.69 to 1.05 over the same range, and needs about 1000 observation times (0.61) to approach the error that the neural schemes reach with 30 (0.38–0.46). The DA error also becomes almost insensitive to the accuracy of the observations (1.40 to 1.46 for observation-error variances from 0.25 to 1.0, against 0.54 to 0.71 at S0), as model error, not observation error, dominates it. The filters still extract some value from additional observations, whereas the hard constraint does not: the RMSE gain of Strong-4DVar from doubling the observations collapses 7.0× under model error, against 1.8–1.9× for the filters, and its error stays on a floor set by the model (1.43 at 30 observations, 1.38 at 1000). Misspecification thus hides most of the information that the observations carry about the state, which supports the statement of Sect. 5.1 that DA-based assessments of an observing system can underestimate its value, all the more under model error.

6 Discussion

Decomposing the error of DA schemes along the chain of Sect. 2, for the mean and for the reported uncertainty, changes how their performance can be read. With a perfect model, the deficit of model-based DA with respect to the best schemes is mostly reducible: on L96, the excess of the ensemble schemes over the best scheme splits into comparable shares of restriction, due to filtering, approximation, due to the ensemble size, and an unattributed term (Sect. 5.1). The first two shrink with smoothing and with more members. The third includes a structural misspecification of the posterior family — Gaussian updates, the mode instead of the mean, a hard constraint — which dominates the budget of the 4D-Var schemes, so that misspecification is not zero even when the dynamical model is exact. This deficit is not calibrated either: the DA ensembles are over-confident, and the more so as their mean improves (Sect. 5.3). Smoothing and larger ensembles thus reduce the error of the mean, not the miscalibration of the reported uncertainty.

Under model error, misspecification dominates the budget of every model-based scheme. On L96, it multiplies their normalised MSE by two to four, and the contrasts that reduced the perfect-model excess no longer help (Sect. 5.2). Representing the uncertainty of the model recovers part of the loss — a weak constraint reduces the error of the hard one by a third to two fifths — without closing the gap. The flow-matching score shows that misspecification degrades the reported distribution even more than the mean: the DA ensembles become more over-confident and short-tailed, and their score at large τ falls below that of climatology, whereas the learned schemes report a distribution whose skill and calibration are unchanged. The advantage of the learned schemes is thus larger for the distribution than for the mean, and the large-τ score reveals misspecification where the mean-based scores only suggest it.

The comparison of priors at matched likelihood (Sect. 5.4) qualifies the common reading that learned schemes win because they learn a better prior. A learned prior outperforms the exact equations used by a 30-member smoother or by a hard constraint, but not by a 100-member smoother: the ordering is set by the approximation budget of the scheme that uses the prior, not

by the fidelity of the prior. Conditioning a learned prior on the model parameters brings little with a perfect model, and becomes a misspecification channel under model error unless the prior is trained with noisy parameters. Composing an amortised mean with a guided sampler gives the best mean in both case studies, but an over-confident distribution, and the composition is what makes the learned prior usable on QG.

Model error also changes what the observations bring (Sect. 5.5). Under model error, additional or more accurate observations bring little to the DA schemes, whose error is set by their model, whereas the learned schemes keep their perfect-model response to the observing system. DA-based assessments of an observing system, such as observing system simulation experiments, may therefore underestimate its value, and the more so as the model is imperfect. Classical DA nevertheless keeps clear strengths. It is better when observations are very sparse, and, together with SDA, it generalises beyond the observing systems seen in training, because it keeps the observation operator and the error covariance explicit: the learned schemes degrade when a window holds fewer or many more observations than in training, and transfer less well between observing systems. On QG, with a realistic observing system, strong-constraint 4D-Var gives the best mean with a perfect model. DA also needs no training set, whereas the learned schemes we compare are trained on simulations of the true system. **[TODO: Back to the puzzle: what the budgets suggest for reanalyses versus observation-only products, stated as hypotheses for real systems.]**

These conclusions come with limitations. The irreducible term $\mathrm{mmse}_p(\tau)$ is unknown, so that every component is a share of an excess over the best scheme, and the cross term between misspecification and approximation is bounded rather than identified. The L96 imperfect-model experiment combines a parametric error with a structural one, as the DA model is truncated to the observed variables **[NEEDS RUN: S1-full rows]**, and the learned rows of this experiment are oracle bounds, since the learned schemes are trained on the true system **[NEEDS RUN: biased-model training arms]**. The flow-matching score is evaluated per element, with sliced evaluations as a check of the dependence structure, and its shape diagnostic has a small negative floor for finite and over-confident ensembles. Finally, the cross-class conclusions rest mostly on L96: on QG, only D-Unet and D-Unet+SDA are available among the learned schemes, and the L63 case study is to be completed.

This study relates to the literature on model error in DA — weak-constraint 4D-Var [45], inflation and model-error estimation [13, 23], assimilation in the unstable subspace [46, 7] — and on learned and hybrid DA [10, 6, 22], including score-based and flow-based DA [43]. **[TODO: Proper scores and their decompositions; mismatched estimation.]** To our knowledge, no controlled study has measured *where* the error of a DA scheme comes from — restriction, misspecification or approximation — for the mean and for the reported uncertainty, at matched information across classical and learned schemes.

Several questions remain open. The first concerns the nature of model error: separating parametric from structural error, as the full-model variant of L96 begins to do, and varying the bias with respect to the stated uncertainty, which is tractable on L63, would show when representing model uncertainty converts misspecification back into restriction. The second concerns the training data: training the learned schemes on the biased model, or on reanalyses, would price the value of simulations of the true system, which our learned schemes use and classical DA does not, and would tell whether their robustness to model error survives without it. The third concerns the scores: joint or sliced versions of the flow-matching score would assess the dependence structure of the reported uncertainty more completely, and its calibration and shape diagnostics could guide the post-processing of ensembles. More broadly, the operator (12) suggests writing every DA scheme as a flow and composing the mean of one with the anomaly of another, as D-Unet+SDA does, with a calibration step to correct the over-confidence of the composition. Finally, the framework extends to the joint estimation of the state and of the model parameters, by including $\boldsymbol{\theta}$ in the target of the flow, which would bring the misspecification of the parameters

into the same budget.

A Case studies

A.1 Lorenz-63

[TODO: Lorenz-63 setup (stochastic Lorenz-63 with wind forcing, parameter randomisation, window length, regular and random observing systems, S0/S1 model errors), to be completed with the hand-ported L63 runs.]

A.2 Lorenz-96

Dynamics. The two-scale system has $K = 8$ slow variables X_k and $J = 4$ fast variables $Y_{j,k}$ per slow variable,

$$\dot{X}_k = X_{k-1}(X_{k+1} - X_{k-2}) - X_k + F - h\sum_j w_j Y_{j,k} + c_1 W(t),$$
$$\dot{Y}_{j,k} = \big[Y_{j+1,k}(Y_{j-1,k} - Y_{j-2,k}) - Y_{j,k} + h_x X_k\big]/\epsilon,$$

with periodic indices, nominal values $F = 8$, $h = h_x = c_1 = 1$, $\epsilon = 0.1$ and coupling weights $w = (1, 1, 0.1, 0.1)$, and a slow external forcing $W(t)$ (an AR(1) process plus a sinusoid). Each window draws its own parameters: F, c_1, h_x, ϵ and w are drawn uniformly within $\pm 20\,\%$ of their nominal values (h is fixed). The system is integrated with a fourth-order Runge–Kutta scheme, $\Delta t = 0.001$, over windows of 3000 steps.

Observations. The observed state has 24 variables: the 8 slow variables and the first two fast variables of each slow variable; the 16 other fast variables are never observed. The observation-error variance is $\mathbf{R} = 0.5\,I$. Two observing systems are used:

- **Regular:** all 24 variables observed at 30 equally spaced times per window (every 100 steps, first step included);
- **Random:** the same windows observed at 10 to 100 times at stratified random positions (first step always observed, at least one observation per DA sub-window), with 4 to 16 of the 16 observable fast variables observed per window.

Tables 2 and 3 and Figs. 1 and 2 use the regular observing system; the random one is analysed in Sect. 5.5. All schemes assimilate exactly the same observations.

Data sets. The test set has 200 windows. The neural schemes are trained on 1000 windows and validated on 100, with a random observing system redrawn at every batch (10 to 300 observation times, 4 to 16 observed fast variables) and fresh observation noise. The DA hyper-parameters are selected on 50 separate validation windows.

Model error. S0 and S1 are described in Sect. 4.1. At S1, the DA model has its parameters biased by $+10\,\%$, receives a corrupted forcing and is truncated to the 24 observed variables; in the S1-full variant, it keeps the 40 variables.

A.3 QG ocean

Dynamics. The two-layer quasi-geostrophic ocean is solved on a 64×64 doubly periodic grid over 1000 km, with a 2 h time step, $\beta = 1.5 \times 10^{-11}\,\mathrm{m^{-1}\,s^{-1}}$ and a layer-depth ratio $\delta = 0.25$. Each window draws its deformation radius (12.75–17.25 km), upper-layer mean flow (0.0425–$0.0575\,\mathrm{m\,s^{-1}}$) and bottom drag. The ocean is forced by the wind-stress curl of a low-order chaotic atmosphere of coupled gyrostats, projected on 12 Fourier modes ($|k| \leq 2$), whose regime changes drive the large-scale circulation; about one test window in five is calm.

Observations. Only the upper-layer streamfunction ψ_1 is observed, on 3 randomly drawn meridional columns per day (4.7 % of the grid), each once per day, with a 5 % observation error, mimicking along-track satellite altimetry. Scores are computed on 100 test windows of 30 days, as the mean over the two streamfunction layers ψ_1, ψ_2.

Model error. At S1, the wind given to the DA model has a +15 % amplitude bias, a random error of 25 % of each mode's standard deviation and a 50 km position error; the deformation radius is biased by −10 % and the bottom drag by −50 %; the observation errors grow to 15 % white plus 15 % correlated per pass; and the DA model runs on a 32×32 grid, which does not resolve the deformation radius.

Data sets. The neural schemes are trained on 5000 windows drawn from the same distribution, with observation-noise augmentation (5–15 % white, 0–15 % correlated per pass) and a random number of observed columns per day. **[TODO: QG training details for D-Unet and SDA (window length, density draw).]**

B Operators and proofs

Gaussian operator. For $q = \mathcal{N}(m, v)$ (per element, standardised units) and $\mathbf{x}_\tau = \tau\mathbf{x}_1 + (1-\tau)\mathbf{x}_0$, the pair $(\mathbf{x}_1, \mathbf{x}_\tau)$ is jointly Gaussian with $\mathrm{Cov}(\mathbf{x}_1, \mathbf{x}_\tau) = \tau v$ and $\mathrm{Var}(\mathbf{x}_\tau) = \tau^2 v + (1-\tau)^2$, so

$$\Psi_q = m + \frac{\tau v}{\tau^2 v + (1-\tau)^2}\,(\mathbf{x}_\tau - \tau m) = m + \frac{\mathrm{snr}\, v}{1 + \mathrm{snr}\, v}\left(\frac{\mathbf{x}_\tau}{\tau} - m\right). \tag{35}$$

Substituting $\mathbf{x}_\tau^\star = \tau\mathbf{x}_1^\star + (1-\tau)\mathbf{x}_0$ and taking the expectation over $\mathbf{x}_0$ gives property (iii).

The chain at every τ. **[TODO: Write out: (a) $\Psi_p(\mathbf{x}_\tau, C) = \mathbb{E}[\mathbf{x}_1^\star \mid \mathbf{x}_\tau, C]$ and $\Psi_{pS} = \mathbb{E}[\mathbf{x}_1^\star \mid \mathbf{x}_\tau, S] = \mathbb{E}[\Psi_p \mid \mathbf{x}_\tau, S]$; (b) the two orthogonality relations; (c) the condition on the scheme's randomness; (d) strict propriety from injectivity of Gaussian smoothing; (e) the $\tau \to 1$ expansion and the $d\mathrm{snr}$ integral (Verdú 2010).]**

Mixture operator. **[TODO: The posterior-weighted average for an empirical or kernel-density output; nearest-member snapping of the atomic form at large τ.]**

C The design matrix

[TODO: Five choices, each a column of Table ??: F1 prior–likelihood split (factorised with explicit $\mathcal{H}$, R vs amortised); F2 dynamics structure (one-step transition, joint window prior, none; equations vs samples); F3 sequential inference (sequential vs whole window; exact only if F2 holds in x and observation errors are white); U-state (none; explicit B, P or ensemble; implicit, learned); U-model (none; state-space Q or inflation; parameter spread; structure). Every scheme's row.]

D Tuning

[TODO: Inflation sweeps of the ETKF, EnKF and ETKS (30 and 100 members) on the validation windows, with the selection rule; Weak-4DVar model-error scale × L-BFGS budget (Table from `docs/results/l96_weak4dvar.md`); SDA guidance weight; D-Unet+SDA start time and guidance.]

E Sensitivities

[TODO: Ensemble size; L-BFGS iterations; training budget and data; flow sampler steps; the 400-epoch rows.]

F The fixed-observing-system protocol and network size

[TODO: From `p1_l96_benchmark.md` annex A: S+ / M / L tiers, VanillaCFM vs PredictStateCFM, $\tau = 0$ heads, SDA conditioning, under the earlier fixed-observing-system protocol; not comparable with the main tables.]

G Estimating FMS_τ

[TODO: Sliced vs per-element evaluation on L96 (snapshot and space–time random projections, 64 directions): same rankings and S0/S1 conclusions; the advantage of SDA over the ETKS at S0 shrinks to 5–8 % on space–time patches.] [TODO: Monte Carlo over $\mathbf{x}_0$ with common random numbers; kernel bandwidth; 24-channel snapshot blocks and their effective-sample-size degeneration at $\tau = 0.75$; window bootstrap.]

H Rank histograms

[TODO: From `docs/results/l96_rank_histograms_c4.md`.]

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